

A Quasi-Static, Curvature-Sourced Modulus-Tracking Mechanism in a 5D Braneworld

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Abstract

We introduce, and study in a 5D braneworld toy model, a quasi-static mechanism by which a compactification modulus tracks a slowly evolving bulk curvature scale. A $\mathbb{Z}_2$ -symmetric 3-brane is embedded in the geometry of a slowly accreting Schwarzschild–Tangherlini black hole, with the accretion rate treated as parametrically small ($\beta \equiv \dot{M}/(MH_0) \ll 1$) and the bulk source as dilute. We work at leading adiabatic order: the bulk is approximated by a sequence of static Schwarzschild–Tangherlini geometries with a slowly drifting mass parameter. The controlled $O(\beta)$ junction calculation (including the exact angular constraint and closed-form accretion corrections) is carried out in Appendix C. The full first-order tracking solution and its numerical integration remain for future work.

The Israel junction conditions yield the standard modified Friedmann constraint, and within the adiabatic approximation we identify a tracking solution $r_b(t) = c(t)R_h(t)$, in which the brane sits at a fixed multiple of the parent horizon while $c(t)$ acquires a slow drift set by the accretion rate and the brane equation of state. A linear stability analysis of the constrained brane sector shows that the tracking branch is an attractor within the zeroth-order generalized expanding window

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho},$$

the familiar $1 < c < \sqrt{2}$ arising only in the vacuum-like case $n_\rho = 0$. For a Goldberger–Wise bulk scalar with brane-localized potentials, we propose that the infrared cutoff of the effective radion potential be set by the local Weyl part of the Kretschmann invariant of the bulk. Combined with the 5D Planck mass this gives a dimensionally correct $[\text{mass}]^4$ cutoff $\Lambda_{\text{IR}} \equiv M_5^2 \sqrt{K_{\text{Weyl}}(r_b, t)} \propto c(t)^{-4} R_h(t)^{-2}$ and, via the dark-dimension scaling $R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}$, a slowly drifting modulus $R_{DD, \text{min}}(t) \propto c(t) R_h(t)^{1/2}$. We stress that the resulting exponent $n = 1/2$ is *not* forced by the geometry: it follows from the specific cutoff choice $\Lambda_{\text{IR}} = M_5^2 \sqrt{K_{\text{Weyl}}}$, whereas the equally dimensionally-admissible choice $\Lambda_{\text{IR}} \propto K_{\text{Weyl}}$ would give $n = 1$. The radion adiabatically follows this slowly moving minimum. In the leading decoupling approximation, the constrained brane mode and radion mode are separately linearly stable when $m_{\text{rad}}^2 > 0$ and the curvature-induced mixing remains perturbative.

We are explicit about the scope of the construction. Self-consistency of the brane Hubble rate pins the bulk horizon to a super-Hubble scale $R_h \sim 1/(\beta H_0)$ and forces $H_b \sim \beta H_0 \ll H_0$: the mechanism operates in a strictly quasi-static regime and does *not* reproduce the observed cosmological expansion. The weak-AdS limit used throughout then requires the nonstandard hierarchy $k \ll \beta H_0$,

inverting the usual Randall–Sundrum relation. The 4D relation $G_4(t) \propto 1/R_{DD}(t)$ and the associated phenomenological estimates rely on a Kaluza–Klein reduction over a compact dark dimension distinct from the noncompact radial direction of the bulk; within the present 5D setup they should be read as the anticipated form of the result in a higher-dimensional (e.g. 6D) completion in which both directions coexist, not as consequences derived here. A full first-order tracking solution (including numerical evolution of the coupled brane–radion system), an explicit higher-dimensional completion, and any extension to a realistic expansion history are left to future work. We present the construction as a proof of concept that bulk curvature can dynamically source a modulus-stabilizing infrared scale, rather than as a complete cosmological model.

1 Introduction

1.1 Limitations of Static Modulus Stabilization

Brane-world constructions in five dimensions have, since their inception, faced the modulus problem: any compactification or warped geometry that introduces a new length scale also introduces a 4D scalar mode whose potential must be generated dynamically. In the original Randall–Sundrum scenarios (1; 2), the inter-brane separation appears as a massless modulus at the classical level; left unstabilized, it leads to a fifth force at macroscopic distances and to an unacceptable time variation of the 4D Newton constant. Goldberger and Wise (3) provided the canonical resolution: a bulk scalar field with quartic brane-localized potentials drives the modulus to a minimum whose location is set by the ratio of brane vacuum expectation values and the bulk scalar mass. Subsequent refinements (4; 5) explored the radion cosmology and finite-temperature effects, the back-reaction of the stabilizing scalar on the warp factor, and the embedding of the mechanism within concrete supergravity completions.

Despite their success, these analyses share a common assumption: the boundary data that determine the modulus potential (the brane tensions, the bulk vacuum, and the infrared cutoff that regulates the contribution of the Casimir or Goldberger–Wise sector) are taken to be strictly time-independent. The minimum of the radion potential is therefore a fixed number, and the only dynamics the radion exhibits is small fluctuation around it. This is appropriate for an idealized empty bulk, but it is restrictive when the bulk contains gravitating sources whose evolution alters the geometry on cosmological time scales. If a higher-dimensional black hole is present in the bulk, its mass parameter sets a curvature scale that the brane sector must register; if that mass parameter evolves, no static stabilization scheme can capture the response.

Several authors (6; 7) have studied moving branes and produced the modified Friedmann equation that follows from the time-dependent Israel junction conditions, including the celebrated dark-radiation term proportional to the bulk-mass parameter divided by r_b^4 . Yet the moduli sector in those treatments is either omitted or reintroduced statically. To our knowledge, no published construction combines (i) a slowly accreting, adiabatically time-dependent bulk geometry, (ii) a moving brane whose dynamics follow from the Israel junction conditions at leading adiabatic order, supplemented by a phenomenological forcing term for the bulk-mass drift, and (iii) a stabilization potential whose infrared cutoff is supplied by the local bulk curvature. Exploring this combination, within a deliberately simplified 5D toy model, is the aim of the present work.

1.2 Dynamic Bulk Geometries and the Dark-Dimension Assumption

We adopt as our working hypothesis the dark-dimension scenario of Montero, Vafa, and Valenzuela (8), in which a single mesoscopic extra dimension of size $R_{DD} \sim 1\text{--}10\ \mu\text{m}$ is compactified on a circle (or equivalent interval) and populated by a Kaluza–Klein tower of gravitons with masses spaced by $1/R_{DD}$. The motivation is structural: combining the swampland distance conjecture (9) with the de Sitter bound leads, under mild assumptions, to a single tower of states becoming light at a scale parametrically related to the infrared cutoff of the 4D effective theory. We take this scenario as given and study its dynamics.

What we add to this picture is a slowly accreting Schwarzschild–Tangherlini black hole in the bulk. The black hole supplies a geometric infrared scale, its horizon radius $R_h(t)$, which evolves on a timescale set by the accretion rate. We characterize that rate by the dimensionless parameter

$$\beta \equiv \frac{\dot{M}}{MH_0}, \quad (1)$$

where M is the physical ADM mass of the bulk black hole. In the 5D AdS–Tangherlini geometry the horizon satisfies $f(R_h) = 0$, i.e. $\mu = R_h^2(1 + k^2 R_h^2)$; in the weak-AdS limit $k^2\mu \ll 1$ adopted throughout this reduces to $R_h^2 \simeq \mu \propto M$. All later relations that use $R_h^2 = \mu$ hold at leading order in $k^2 R_h^2$ and are not exact. Consequently, $\dot{\mu}/\mu = \dot{M}/M$, so the same parameter may equivalently be written as $\beta \equiv \dot{\mu}/(\mu H_0)$ throughout the derivations that follow. We work throughout in the regime $\beta \ll 1$, in which the bulk is well-approximated by a sequence of Schwarzschild–Tangherlini geometries connected by a slowly varying mass parameter. The dark-dimension assumption fixes the swampland scaling relation

$$R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}, \quad (2)$$

where Λ_{IR} is the 4D vacuum-energy scale (dimensions $[\text{mass}]^4$) of the 4D effective theory. When we couple this relation to the curvature-set Λ_{IR} derived in Section 4, the dynamical exponent $n = 1/2$ follows directly, given the minimal cutoff choice $\Lambda_{\text{IR}} = M_5^2 \sqrt{K_{\text{Weyl}}}$ (the naive alternative $\Lambda_{\text{IR}} \propto K_{\text{Weyl}}$ would instead give $n = 1$; see Section 4).

1.3 Summary of Main Results

Within this deliberately simplified 5D toy model, we present three main results, each stated with its scope and caveats:

1. The brane position $r_b(t)$ adiabatically tracks the parent horizon $R_h(t)$ via the relation $r_b(t) = c(t)R_h(t)$ (with slowly varying $c(t)$), as follows from the Israel junction conditions at leading adiabatic order, with the brane equation of motion derived at leading order from the Israel junction conditions (controlled $O(\beta)$ corrections given in Appendix C). This scaling causes the bulk-induced dark-radiation term to scale together with the brane spatial-curvature contribution up to the slow drift of $c(t)$.
2. The Weyl part of the Kretschmann invariant of the bulk geometry, combined with the 5D Planck mass to make a dimensionally correct $[\text{mass}]^4$ object, supplies a curvature-set infrared cutoff $\Lambda_{\text{IR}} \propto c(t)^{-4} R_h(t)^{-2}$ (under the minimality assumption that the AdS scale k is not included in the cutoff construction), which, via the dark-dimension scaling (2), yields the dynamical compactification radius $R_{DD,\text{min}}(t) \propto$

$c(t)R_h(t)^{1/2}$ (Section 4). The exponent $n = 1/2$ follows from this construction; it is contingent on the cutoff choice $\Lambda_{\text{IR}} = M_5^2 \sqrt{K_{\text{Weyl}}}$ (curvature power $b = 1/2$) and is *not* forced by the geometry alone, since the alternative dimensionally-consistent choice $\Lambda_{\text{IR}} \propto K_{\text{Weyl}}$ ($b = 1$) would give $n = 1$.

3. The radion field adiabatically follows this moving minimum. In a leading decoupling approximation, a linear stability analysis shows that the constrained brane mode and the radion mode are separately stable inside the generalized high-energy tracking window

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho},$$

provided the radion mass satisfies $m_{\text{rad}}^2 > 0$ and the curvature-induced brane–radion mixing remains perturbative. The familiar window $1 < c < \sqrt{2}$ is recovered only in the special vacuum-like case $n_\rho = 0$. We emphasize that this regime is strictly quasi-static: self-consistency of the brane Hubble rate fixes the bulk horizon to a super-Hubble scale $R_h \sim 1/(\beta H_0)$ and forces $H_b \sim \beta H_0 \ll H_0$, so the construction does *not* describe the observed cosmological expansion (Section 5). The associated 4D relation $G_4(t) \propto 1/R_{DD}(t)$ follows from a Kaluza–Klein reduction over a compact dark dimension that is distinct from the noncompact radial direction of the bulk; within the present 5D setup it is therefore the anticipated form of the result in a higher-dimensional (e.g. 6D) completion, rather than a consequence derived here.

2 The 5D Setup

2.1 5D Einstein–Hilbert Action with Bulk Matter

We start from the 5D Einstein–Hilbert action with a bulk cosmological constant $\Lambda_5 < 0$ (anti-de Sitter bulk), coupled to a dilute bulk matter sector and a 3-brane:

$$S = \frac{M_5^3}{2} \int d^5x \sqrt{-g^{(5)}} (R^{(5)} - 2\Lambda_5) + \int d^5x \sqrt{-g^{(5)}} \mathcal{L}_{\text{bulk}} + S_{\text{brane}}, \quad (3)$$

with M_5 the 5D Planck mass and $\kappa_5^2 \equiv M_5^{-3}$ the corresponding gravitational coupling. In our conventions κ_5^2 has dimensions $[\text{length}]^3 = [\text{mass}]^{-3}$ and σ (the brane tension, dimensions $[\text{mass}]^4$) appears in the brane action as a 4D energy density. We impose the Randall–Sundrum fine-tuning condition

$$\Lambda_5 = -\frac{\kappa_5^4 \sigma^2}{6}, \quad (4)$$

which has dimensions $[\text{mass}]^2$, matching Λ_5 . With $k^2 \equiv -\Lambda_5/6 = \kappa_5^4 \sigma^2/36$, the AdS-curvature scale and the brane tension are related by

$$k = \frac{\kappa_5^2 \sigma}{6}. \quad (5)$$

This relation exactly cancels the effective 4D cosmological constant generated by the brane tension, ensuring that the only remaining contributions to the brane cosmology are matter, spatial curvature, and dark radiation.

The bulk Lagrangian $\mathcal{L}_{\text{bulk}}$ describes a dilute null-dust source whose stress-energy takes the Vaidya-like form

$$T_{ab}^{\text{bulk}} = \rho_{\text{acc}}(v, r) k_a k_b, \quad k^a k_a = 0, \quad (6)$$

with v an ingoing null coordinate. This choice reproduces the 5D analogue of the 4D Vaidya solution and yields, on integration of the radial component of the Einstein equation,

$$\frac{d\mu(t)}{dv} = \mathcal{C}_5 \kappa_5^2 r^3 \rho_{\text{acc}}(v, r) \Big|_{r \rightarrow \infty}, \quad \mathcal{C}_5 = O(1), \quad (7)$$

where $\mathcal{C}_5$ is fixed by the normalization of the null vector k_a and the $D = 5$ geometry (in the standard 5D Vaidya normalization $\mathcal{C}_5 = 2/3$), so that $\mu(t)$ grows monotonically when the source has positive energy density. We require the slow-accretion bound $|(d\mu/dt)/\mu| \ll H_0$, which justifies treating the geometry as a slowly varying sequence of static Schwarzschild–Tangherlini backgrounds at zeroth adiabatic order. This treatment is performed at leading adiabatic order. The controlled $O(\beta)$ junction calculation is given in Appendix C; the full first-order tracking solution and its numerical integration remain for future work.

2.2 Schwarzschild–Tangherlini Bulk with Time-Dependent Mass

Away from the brane, we work at **leading (zeroth) order in the slow-accretion parameter** $\beta \ll 1$. At this order the bulk geometry is described by the diagonal 5D AdS–Schwarzschild–Tangherlini metric

$$ds_{(5)}^2 = -f(r, t) dt^2 + f(r, t)^{-1} dr^2 + r^2 d\Omega_3^2, \quad f(r, t) = 1 - \frac{\mu(t)}{r^2} + k^2 r^2. \quad (8)$$

Here $d\Omega_3^2$ is the metric on the unit three-sphere and $\mu(t)$ has dimensions [length]².

This diagonal form is **not** an exact solution of the Einstein equations when $\mu(t)$ is time-dependent. A null-dust source produces off-diagonal terms, in particular a G_r^t component. However, at leading order in $\beta \ll 1$ we drop all $O(\beta)$ corrections to the metric functions themselves. The explicit accretion term $\partial_t \mu$ is itself $O(\beta)$; at strictly zeroth adiabatic order it is dropped from the brane equation of motion (Section 3), and its controlled $O(\beta)$ form is derived from the ingoing-Vaidya bulk in Appendix C.

Consequently, the treatment is **not** a uniformly controlled expansion to first order in β . It should be understood as a leading-order treatment that uses the simpler static geometry for the extrinsic curvature; the leading effect of slow accretion enters the brane equation of motion only at $O(\beta)$, and is treated in the controlled Vaidya calculation of Appendix C. The controlled $O(\beta)$ junction calculation is given in Appendix C; the full first-order tracking solution (including consistent back-reaction on the brane) remains for future numerical work.

A useful invariant of (8) is the Kretschmann scalar, which decomposes into a constant AdS piece and a Weyl (Schwarzschild) piece:

$$K(r, t) = R_{abcd} R^{abcd} = 40k^4 + \frac{72\mu(t)^2}{r^8}. \quad (9)$$

The constant AdS background $40k^4$ does not source non-trivial radial dependence and is absorbed into the bulk cosmological-constant counterterm. The physically relevant

curvature signal from the black hole is the Weyl part:

$$K_{\text{Weyl}}(r, t) \equiv C_{abcd}C^{abcd} = \frac{72\mu(t)^2}{r^8}. \quad (10)$$

Equation (10) is the central geometric input for the curvature-set infrared cutoff of Section 4.

2.3 Brane Content and Goldberger–Wise Bulk Scalar

The 3-brane is taken to be $\mathbb{Z}_2$ -symmetric and to carry tension σ , four-dimensional brane matter (whose equation of state is constrained to $w < -2/3$ by the tracking analysis of Section 3.3, so it is dark-energy-like rather than ordinary dust or radiation), and brane-localized potentials for a bulk scalar Φ :

$$S_{\text{brane}} = \int_{\Sigma} d^4x \sqrt{-h} \left(-\sigma + \mathcal{L}_{\text{matter}}(h_{\mu\nu}, \Psi) - V_{\text{br}}(\Phi) \right), \quad (11)$$

with $h_{\mu\nu}$ the induced metric on the brane worldvolume Σ . The bulk scalar action is

$$S_{\Phi} = - \int d^5x \sqrt{-g^{(5)}} \left[\frac{1}{2} (\nabla\Phi)^2 + V_{\text{bulk}}(\Phi) \right], \quad (12)$$

and the brane-localized potential is taken in the canonical Goldberger–Wise quartic form,

$$V_{\text{br}}(\Phi) = \frac{\lambda_{\text{br}}}{4} (\Phi^2 - v_{\text{br}}^2)^2, \quad V_{\text{bulk}}(\Phi) = \frac{1}{2} m_{\text{GW}}^2 \Phi^2. \quad (13)$$

In the standard two-brane Goldberger–Wise mechanism the modulus is fixed by boundary conditions at both ends of the extra dimension. Here we employ the mechanism *phenomenologically*: we retain only the visible-brane potential and treat the resulting effective radion potential as given, with the infrared cutoff Λ_{IR} supplied dynamically by the bulk curvature (see Section 4). The radion of the 4D effective theory is identified with the canonically normalized fluctuation $\phi(x)$ of the compactification radius,

$$R_{DD}(x) = R_0 e^{\phi(x)/f_{\phi}}, \quad (14)$$

where ϕ is distinct from the bulk scalar Φ , R_0 is a fiducial reference scale, and $f_{\phi} \sim M_{\text{Pl}}$ is fixed by the dimensional reduction. We stress at the outset that the compact dark dimension hosting R_{DD} is logically distinct from the noncompact radial direction r of the bulk black hole: the present construction couples a five-dimensional curvature-sourcing sector to a compact dark-dimension modulus sector, and a fully geometric realization in which both coexist requires a higher-dimensional (e.g. 6D) completion (see Section 5). The effective potential for the radion inherits three contributions:

$$V_{\text{eff}}(\phi; M(t)) = V_{\text{Cas}}(\phi) + V_{\text{GW}}(\phi; \Lambda_{\text{IR}}(M(t))) + V_{\text{curv}}(\phi; K_{\text{Weyl}}(r_b, t)), \quad (15)$$

with the explicit form of each term derived in Section 4. The non-trivial step is the appearance of the time-dependent infrared cutoff $\Lambda_{\text{IR}}(M(t))$ inside the Goldberger–Wise contribution; this links the modulus sector to the bulk accretion history and yields the $n = 1/2$ scaling of Section 4, under the cutoff choice adopted there.

3 Leading-Order Time-Dependent Israel Junction Conditions and Brane Equation of Motion

3.1 Induced Metric and Junction Conditions

We embed the brane at $r_b(\tau)$ with induced metric

$$ds_{(4)}^2 = -d\tau^2 + r_b(\tau)^2 d\Omega_3^2, \quad (16)$$

where τ is the proper time on the brane. Throughout the rest of this section overdots denote derivatives with respect to τ , i.e. $\dot{r}_b \equiv dr_b/d\tau$. We emphasize that proper and coordinate time are *not* interchangeable in the regime of interest: for slow brane motion the redshift factor is $\dot{t} \simeq 1/\sqrt{f(r_b)} = (1 - c^{-2})^{-1/2}$ in the weak-AdS limit, which diverges as $c \rightarrow 1^+$ and equals $\sqrt{2}$ at $c = \sqrt{2}$. It is never close to unity inside the allowed tracking window, especially near its lower endpoint $c \rightarrow 1^+$, and every conversion between coordinate-time and proper-time rates below retains this factor explicitly. The brane four-velocity and outward unit normal in the bulk coordinates of (8) are

$$u^a = (\dot{t}, \dot{r}_b, 0, 0, 0), \quad \dot{t} = \frac{\sqrt{f(r_b, t) + \dot{r}_b^2}}{f(r_b, t)}, \quad (17)$$

$$n^a = \left(\frac{\dot{r}_b}{f(r_b, t)}, \sqrt{f(r_b, t) + \dot{r}_b^2}, 0, 0, 0 \right), \quad (18)$$

satisfying $u^a u_a = -1$, $n^a n_a = +1$, $n^a u_a = 0$ (these are easily verified directly).

Imposing $\mathbb{Z}_2$ symmetry, the Israel junction conditions (10) read

$$[K_{\mu\nu}] = -\kappa_5^2 \left(S_{\mu\nu} - \frac{1}{3} S h_{\mu\nu} \right). \quad (19)$$

A direct computation (recorded in Appendix A) yields the extrinsic-curvature components on the moving brane:

$$K_j^i(r_b) = \frac{\sqrt{f(r_b, t) + \dot{r}_b^2}}{r_b} \delta_j^i, \quad (20)$$

$$K_\tau^\tau(r_b) = \frac{\ddot{r}_b + \frac{1}{2} \partial_r f(r_b, t)}{\sqrt{f(r_b, t) + \dot{r}_b^2}} = \frac{\ddot{r}_b + \mu(t)/r_b^3 + k^2 r_b}{\sqrt{f(r_b, t) + \dot{r}_b^2}}, \quad (21)$$

where in the second form we used $\partial_r f = 2\mu/r^3 + 2k^2 r$. The angular projection of the Israel condition is now known to be exact (even in the ingoing-Vaidya bulk), while the brane equation of motion below is treated at leading adiabatic order, approximating the extrinsic curvature by its static- f form. The controlled $O(\beta)$ corrections to the $\tau\tau$ junction and to brane-bulk energy exchange are derived in Appendix C; the full first-order tracking problem remains for future numerical study. Both expressions reduce to the standard Binétruy–Deffayet–Ellwanger–Langlois results in the static limit $\partial_t \mu = 0$.

For a brane carrying tension σ and a perfect-fluid matter content with energy density ρ_{br} and pressure p_{br} ,

$$S^\mu{}_\nu = \text{diag}(-\sigma - \rho_{\text{br}}, -\sigma + p_{\text{br}}, -\sigma + p_{\text{br}}, -\sigma + p_{\text{br}}), \quad (22)$$

with trace $S = -4\sigma - \rho_{\text{br}} + 3p_{\text{br}}$. Using the $\mathbb{Z}_2$ relation $[K_{\mu\nu}] = 2K_{\mu\nu}$, the angular projection of (19) yields

$$\frac{\sqrt{f(r_b, t) + \dot{r}_b^2}}{r_b} = \frac{\kappa_5^2}{6}(\sigma + \rho_{\text{br}}), \quad (23)$$

(up to an overall sign that is absorbed into the choice of normal direction; only the squared relation is used below). Squaring and rearranging gives

$$\dot{r}_b^2 = -f(r_b, t) + r_b^2 \left(\frac{\kappa_5^2(\sigma + \rho_{\text{br}})}{6} \right)^2. \quad (24)$$

Substituting $f = 1 - \mu/r_b^2 + k^2 r_b^2$, expanding $(\sigma + \rho_{\text{br}})^2 = \sigma^2 + 2\sigma\rho_{\text{br}} + \rho_{\text{br}}^2$, and using the RS relation (4) to cancel the $+k^2 r_b^2$ term in f against the $r_b^2(\kappa_5^2\sigma/6)^2 = k^2 r_b^2$ term, we obtain the modified Friedmann equation:

$$H_b^2 = \frac{\kappa_5^4}{36}\rho_{\text{br}}^2 + \frac{\kappa_5^4\sigma\rho_{\text{br}}}{18} + \frac{\mu(t)}{r_b^4} - \frac{1}{r_b^2}. \quad (25)$$

The four terms are, in order, the quadratic ρ_{br}^2 brane-world contribution, the linear $\sigma\rho_{\text{br}}$ term reproducing standard four-dimensional cosmology (subdominant in the high-energy regime $\rho_{\text{br}} \gg \sigma$ adopted below), the dark-radiation contribution sourced by the bulk black hole, and the spatial-curvature contribution induced by the embedding.

Key cancellation. Note explicitly that the RS tuning has been used: the $k^2 r_b^2$ term from f in (24) cancels exactly against the brane-tension piece $(\kappa_5^2\sigma/6)^2 r_b^2 = k^2 r_b^2$, leaving no residual $k^2 r_b^2$ contribution in (25). This cancellation is essential for the next subsection, where the effective brane potential is constructed.

3.2 Brane Equation of Motion

The constraint (24) can be rewritten, in the form of an energy equation for the brane radius treated as a mechanical degree of freedom,

$$\frac{1}{2}\dot{r}_b^2 + V_{\text{eff}}^{\text{brane}}(r_b, t) = -\frac{1}{2}, \quad (26)$$

where, after the RS cancellation noted above, the effective potential is

$$V_{\text{eff}}^{\text{brane}}(r_b, t) = -\frac{\mu(t)}{2r_b^2} - \frac{\kappa_5^4\sigma\rho_{\text{br}}r_b^2}{36} - \frac{\kappa_5^4\rho_{\text{br}}^2r_b^2}{72}. \quad (27)$$

Crucially, there is no $k^2 r_b^2$ term in $V_{\text{eff}}^{\text{brane}}$: the cancellation between the AdS contribution from f and the tension contribution from $(\sigma + \rho_{\text{br}})^2$ is exact. The second-order brane equation of motion is obtained by differentiating the Friedmann constraint (25) along the brane worldline and imposing stress-energy conservation; by the contracted Gauss–Codazzi identity this is equivalent to the $\tau\tau$ -component of the Israel condition (19). Because the brane matter is conserved ($\rho_{\text{br}} \propto r_b^{-n_\rho}$), the matter terms must be differentiated *along that trajectory*: it is the total derivative with respect to r_b , not the partial derivative at fixed ρ_{br} . the resulting high-energy acceleration is given in Section 3.3, Eq. (37), and carries the full equation-of-state dependence—the n_ρ -dependent term survives even at $\dot{r}_b = 0$ and fixes the force balance (40). Working consistently in proper time τ (in which the constraint is naturally expressed), the redshift between coordinate

and proper time is already incorporated through $\dot{t}$, and no separate velocity-dependent friction factor appears.

At strictly zeroth adiabatic order the explicit accretion force is $O(\beta)$ and is dropped. The controlled $O(\beta)$ correction is *not* obtained by inserting a term $+\partial_t \mu \dot{t}/(2r_b^2)$ into the diagonal static metric. The correct forcing, derived from the ingoing-Vaidya bulk in Appendix C, is

$$\ddot{r}_b = \ddot{r}_b^{\text{static}} - \frac{\mu'(v) \dot{v}^2}{2r_b^2}, \quad \dot{v} \equiv \frac{dv}{d\tau}, \quad (28)$$

which is $O(\beta)$, of the same order as $\ddot{c}$, $\dot{c} \dot{R}_h$, $\ddot{R}_h$, and the brane–bulk energy-exchange source; a genuinely first-order tracking solution must retain all of these simultaneously. The tracking analysis below is therefore a zeroth-order adiabatic force-balance analysis. The redshift factor $\dot{t}$ converting coordinate-time to proper-time derivatives (see Appendix A) satisfies, for slow brane motion, $\dot{t} \simeq 1/\sqrt{f(r_b)} = (1 - c^{-2})^{-1/2}$ in the weak-AdS limit; this is of order unity only for c well above 1 and *diverges* as $c \rightarrow 1^+$, so it cannot be set to unity inside the window $1 < c < \sqrt{2}$, and is retained explicitly wherever it appears. The stability of the tracking branch is *not* a friction effect; it is established from the constrained Friedmann perturbation analysis of Appendix B, where the single physical brane mode (after the first-order Israel constraint is imposed) has eigenvalue $-H_b|_0$ inside the generalized expanding window.

3.3 The Adiabatic-Tracking Solution

We now establish that the scaling relation

$$r_b(t) = c(t) R_h(t), \quad (29)$$

with slowly varying $c(t)$, is an adiabatic tracking solution of the brane dynamics in the slow-accretion regime $\beta \ll 1$.

Differentiating the ansatz with respect to proper time τ gives the exact kinematic relations

$$\dot{r}_b = \dot{c} R_h + c \dot{R}_h, \quad (30)$$

$$\ddot{r}_b = \ddot{c} R_h + 2\dot{c} \dot{R}_h + c \ddot{R}_h. \quad (31)$$

From the horizon evolution with respect to coordinate time t , we have

$$\frac{1}{R_h} \frac{dR_h}{dt} = \frac{1}{2} \beta H_0. \quad (32)$$

Converting to brane proper time τ through

$$\dot{t} = \frac{\sqrt{f(r_b, t) + \dot{r}_b^2}}{f(r_b, t)} \quad (33)$$

yields

$$\frac{1}{R_h} \frac{dR_h}{d\tau} = \dot{t} \frac{1}{2} \beta H_0. \quad (34)$$

At the brane the redshift factor that converts coordinate-time derivatives to proper time is

$$\dot{t} = \frac{\sqrt{f(r_b) + \dot{r}_b^2}}{f(r_b)}.$$

On the background tracking solution this factor is $O(1)$ throughout the interior of the allowed window and only diverges as $c \rightarrow 1^+$. The simple slow-motion approximation $\dot{t} \approx (1 - c^{-2})^{-1/2}$ is therefore not uniformly valid, but the difference is an $O(1)$ coefficient that does not affect the leading parametric estimates

$$R_h \sim \frac{1}{\beta H_0}, \quad H_b \sim \beta H_0.$$

We retain the exact $\dot{t}$ wherever it appears in drift-rate expressions, but for all scaling arguments it is sufficient to know that $\dot{t} = O(1)$ away from the lower boundary of the window.

The accretion-driven piece in proper time is therefore

$$\frac{\dot{R}_h}{R_h} = \dot{t} \frac{1}{2} \beta H_0, \quad (35)$$

which remains $O(\beta H_0)$ provided c is bounded away from 1.

Quasi-static tracking condition. We obtain the leading tracking condition by imposing quasi-static balance, $\ddot{r}_b = 0$, on the brane acceleration. We caution, however, that the brane response scale on the tracking solution is itself $H_b|_0 = O(\beta H_0)$ (Eq. (55)), i.e. of the *same* order as the accretion-driven drift $\dot{c}/c = O(\beta H_0)$. Consequently the kinematic terms $\ddot{c}$, $\dot{c} \dot{R}_h$, and $\ddot{R}_h$ enter $\ddot{r}_b/r_b$ at $O(\beta^2 H_0^2)$ —the same order as the retained force-balance terms $(1 - n_\rho)B$ and μ/r_b^4 , as confirmed in Section 6 and Appendix C—and are *not* suppressed by an independent adiabatic parameter. Imposing $\ddot{r}_b = 0$ is therefore a leading-order *parametric* balance that fixes $c = O(1)$ and the scaling of the tracking relation, but not its coefficient: the neglected same-order terms shift the coefficient in (40), and the endpoints of the window, by an order-unity relative amount, deferred to the first-order problem of Appendix C.

A subtle point is important. Because the brane matter is conserved ($\rho_{\text{br}} \propto r_b^{-n_\rho}$), the force balance must employ the *total* derivative of the effective potential along that trajectory. the resulting high-energy acceleration (whose quasi-static balance yields the tracking condition) is given in Eq. (37).

To obtain the consistent high-energy balance, differentiate the high-energy Friedmann constraint

$$H_b^2 = B + \frac{\mu}{r_b^4} - \frac{1}{r_b^2}, \quad B \equiv \frac{\kappa_5^4 \rho_{\text{br}}^2}{36}, \quad (36)$$

with $\rho_{\text{br}} \propto r_b^{-n_\rho}$. At zeroth adiabatic order, neglecting $\dot{\mu}$, this gives

$$\frac{\ddot{r}_b}{r_b} = (1 - n_\rho)B - \frac{\mu}{r_b^4}. \quad (37)$$

The quasi-static condition $\ddot{r}_b = 0$, together with $r_b = cR_h$ and $\mu = R_h^2$, gives

$$\frac{1}{c^4 R_h^2} = (1 - n_\rho)B. \quad (38)$$

Equivalently,

$$B = \frac{1}{(1 - n_\rho) c^4 R_h^2}. \quad (39)$$

Using $B = \kappa_5^4 \rho_{\text{br}}^2 / 36$, the tracking relation becomes

$$c^4(t) = \frac{36}{(1 - n_\rho) \kappa_5^4 R_h^2(t) \rho_{\text{br}}^2(t)}. \quad (40)$$

This reduces to the naive fixed- ρ_{br} force-balance result only in the formal case $n_\rho = 0$. For $B > 0$, the high-energy tracking root requires

$$n_\rho < 1. \quad (41)$$

For conserved FRW matter $\rho_{\text{br}} \propto a^{-3(1+w)}$ with $a = r_b$ we have $n_\rho = 3(1+w)$, so the tracking branch requires $w < -2/3$: only vacuum- or dark-energy-like matter is admissible. In particular ordinary dust ($w = 0$, $n_\rho = 3$) and radiation ($w = \frac{1}{3}$, $n_\rho = 4$) lie *outside* the allowed window and do not support a positive high-energy tracking root in this simplified force-balance model; such cases require additional sources, bulk-brane energy exchange, or a separate treatment.

The tracking relation (40) determines $c(t)$ from the instantaneous values of $R_h(t)$ and $\rho_{\text{br}}(t)$. We restrict to the physically preferred branch

$$c(t) > 1, \quad (42)$$

so that the brane lies outside the parent horizon.

Constrained stability of the tracking branch. The Israel junction condition (25) is a first-order constraint, so the brane sector contains only one physical perturbative mode after the constraint is imposed. Perturb around the tracking solution by writing

$$r_b(t) = c(t) R_h(t) (1 + \varepsilon(t)), \quad |\varepsilon| \ll 1. \quad (43)$$

Including the matter-density perturbation

$$\frac{\delta \rho_{\text{br}}}{\rho_{\text{br}}} = -n_\rho \varepsilon, \quad (44)$$

the linearized Friedmann constraint gives

$$\dot{\varepsilon} = -H_b|_0 \varepsilon + O(\beta H_0 \varepsilon). \quad (45)$$

The background Hubble rate on the tracking solution is

$$H_b^2|_0 = \frac{1}{c^4 R_h^2} \left(\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right). \quad (46)$$

Therefore the constrained brane perturbation decays whenever the tracking background is expanding. For $B > 0$, this requires

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}. \quad (47)$$

The special vacuum-like case $n_\rho = 0$ reduces to

$$1 < c < \sqrt{2}. \quad (48)$$

Thus $1 < c < \sqrt{2}$ is not the generic stability window; it is the $n_\rho = 0$ limit of the generalized high-energy window (47).

Note on tracking vs. cosmological expansion. The drift of the embedding ratio on the tracking solution is

$$\frac{\dot{c}}{c} = \frac{n_\rho - 1}{2 - n_\rho} \frac{\dot{R}_h}{R_h}. \quad (49)$$

The drift of $c(t)$ is fixed self-consistently by the tracking relation (40) together with brane stress-energy conservation. Since the high-energy tracking relation implies

$$c^4 R_h^2 \rho_{\text{br}}^2 = \text{const.}, \quad (50)$$

differentiating gives

$$4 \frac{\dot{c}}{c} + 2 \frac{\dot{R}_h}{R_h} + 2 \frac{\dot{\rho}_{\text{br}}}{\rho_{\text{br}}} = 0. \quad (51)$$

Using

$$\frac{\dot{\rho}_{\text{br}}}{\rho_{\text{br}}} = -n_\rho H_b = -n_\rho \left(\frac{\dot{c}}{c} + \frac{\dot{R}_h}{R_h} \right), \quad (52)$$

one obtains

$$\frac{\dot{c}}{c} = \frac{n_\rho - 1}{2 - n_\rho} \frac{\dot{R}_h}{R_h}. \quad (53)$$

Together with (34), this becomes

$$\frac{\dot{c}}{c} = \frac{n_\rho - 1}{2 - n_\rho} \frac{\dot{R}_h}{R_h} = \frac{n_\rho - 1}{2(2 - n_\rho)} \dot{t} \beta H_0 = O(\dot{t} \beta H_0). \quad (54)$$

Consequently,

$$H_b = \frac{\dot{c}}{c} + \frac{\dot{R}_h}{R_h} = \frac{1}{2 - n_\rho} \frac{\dot{R}_h}{R_h} = \frac{\dot{t} \beta H_0}{2(2 - n_\rho)}. \quad (55)$$

Consistency of the two expressions for H_b . The angular junction condition gives (46), while the kinematic tracking relation gives (55). These agree only if the bulk scale is fixed by

$$R_h = \frac{2(2 - n_\rho)}{c^2 \dot{t} \beta H_0} \left[\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right]^{1/2}. \quad (56)$$

If one further imposes the strict slow-radial-motion sublimit $\dot{r}_b^2 \ll f$, then

$$\dot{t} \simeq \frac{c}{\sqrt{c^2 - 1}}, \quad (57)$$

and Eq. (56) reduces to

$$R_h = \frac{2(2 - n_\rho) \sqrt{c^2 - 1}}{c^3 \beta H_0} \left[\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right]^{1/2}. \quad (58)$$

This form should be read only as the slow-motion approximation. On the tracking solution, however, $\dot{r}_b^2 = H_b^2 r_b^2 = \frac{1}{c^2} \left(\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right)$ is generically $O(1)$ inside the window, and keeping it in the weak-AdS limit gives instead

$$\dot{t} = \frac{c}{(c^2 - 1) \sqrt{1 - n_\rho}}, \quad (59)$$

so the exact numerical prefactor in R_h differs, although the parametric result $R_h \sim (\beta H_0)^{-1}$ is unchanged away from the endpoints of the window.

Thus R_h is not a free parameter. At leading order in β , consistency pins it to a scale of order

$$R_h \sim \frac{1}{\beta H_0}. \quad (60)$$

The weak-AdS condition $k^2 R_h^2 \ll 1$ together with this pinning then requires

$$k \ll \frac{1}{R_h} \sim \beta H_0, \quad (61)$$

inverting the usual Randall–Sundrum hierarchy. Reality of R_h , together with $c > 1$ and positive $\sigma \rho_{\text{br}}$, requires exactly the generalized condition

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}. \quad (62)$$

The earlier $1 < c < \sqrt{2}$ result is recovered only when $n_\rho = 0$.

The clean quasi-static estimates

$$R_h \sim \frac{1}{\beta H_0}, \quad H_b \sim \beta H_0, \quad \dot{t} = O(1) \quad (63)$$

therefore refer only to the interior of the generalized window, bounded away from its endpoints. They additionally require n_ρ to be bounded away from 1, since $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}} \propto (1-n_\rho)^{-1/2}$ diverges as $n_\rho \rightarrow 1^-$. Near the lower endpoint $c \rightarrow 1^+$, the redshift factor $\dot{t}$ diverges. Near the upper endpoint of the generalized window, the square root in (56) vanishes. In either limit, the simple quasi-static estimates break down. The mismatch between the conservation-fixed drift (54) and the time derivative of (56) is $O(\beta^2 H_0^2)$, beyond the leading adiabatic order retained here.

4 Modulus Potential with Curvature-Set Infrared Cutoff

4.1 Effective Radion Potential

Dimensional reduction of (3)–(13) on the compactified bulk yields the 4D effective potential (68). We here derive each contribution, identifying the Λ_{IR} -dependence as the source of explicit time variation at fixed radion.

Casimir contribution. Quantizing the bulk graviton on the compact circle of radius R_{DD} , the standard Coleman–Weinberg integration over the Kaluza–Klein tower gives

$$V_{\text{Cas}}(\phi) = -\frac{c_{\text{Cas}}}{R_{DD}^4(\phi)}, \quad c_{\text{Cas}} > 0, \quad (64)$$

with c_{Cas} a numerical coefficient encoding the bosonic helicities of the bulk graviton and the topology. The dependence $V_{\text{Cas}} \propto R_{DD}^{-4}$ reflects pure dimensional analysis in 4D units. At fixed ϕ , (64) has no explicit time dependence.

Goldberger–Wise contribution. The bulk scalar Φ satisfies its 5D equation of motion with the brane-localized boundary potential (13). In the canonical $\lambda_{\text{br}} \rightarrow \infty$

limit, $\Phi|_{\text{brane}} = v_{\text{br}}$. We take the contribution of the Goldberger–Wise sector to the effective 4D radion potential to be of the phenomenological form

$$V_{\text{GW}}(\phi; \Lambda_{\text{IR}}) = \Lambda_{\text{IR}} \mathcal{F}_{\text{GW}} \left(\frac{v_{\text{br}}}{M_5^{3/2}}, m_{\text{GW}} R_{DD} \right), \quad (65)$$

where Λ_{IR} has dimensions $[\text{mass}]^4$ (an energy-density-scale cutoff) and $\mathcal{F}_{\text{GW}}$ is a dimensionless profile function. The linear dependence on Λ_{IR} is consistent with dimensional analysis. While other combinations involving higher powers of Λ_{IR} or additional mass scales are possible, the linear construction provides the most minimal setup compatible with the scaling requirements of the model. The explicit form of $\mathcal{F}_{\text{GW}}$ is known analytically in the limit $m_{\text{GW}} R_{DD} \ll 1$, where it produces the celebrated logarithmic hierarchy generation. In the present dynamical setting the cutoff $\Lambda_{\text{IR}}(t)$ becomes time-dependent through the bulk curvature evaluated at the moving brane.

Curvature contribution. The bulk scalar admits a non-minimal coupling to curvature. Because the leading curvature dependence arises from the Weyl tensor, we include the dimensionally consistent operator

$$\Delta \mathcal{L}_5 \supset \frac{\xi}{M_5^2} K_{\text{Weyl}}(r_b, t) \Phi^2, \quad (66)$$

where the factor of M_5^{-2} ensures the operator has mass dimension five in the 5D theory. Upon dimensional reduction over the compact direction ($\int dy \rightarrow R_{DD}$, with Φ the bulk Goldberger–Wise scalar and ϕ the 4D radion) this produces the contribution

$$V_{\text{curv}}(\phi; K_{\text{Weyl}}(r_b, t)) = \frac{\xi}{M_5^2} K_{\text{Weyl}}(r_b, t) \langle \Phi^2 \rangle(\phi), \quad (67)$$

where $\langle \Phi^2 \rangle(\phi)$ denotes the *dimensionally-reduced* expectation value of mass dimension two: the factor $\int dy = R_{DD}$ from the compactification integral has been absorbed into its definition, so that V_{curv} carries the correct $[\text{mass}]^4$ dimensions of a 4D energy density. (In bulk variables $\langle \Phi^2 \rangle_5$ has dimension three and the explicit R_{DD} appears: $V_{\text{curv}} = \xi M_5^{-2} R_{DD} K_{\text{Weyl}} \langle \Phi^2 \rangle_5$.) **Total effective potential.** Collecting the Casimir, Goldberger–Wise, and curvature contributions gives the full effective radion potential

$$V_{\text{eff}}(\phi; M(t)) = -\frac{c_{\text{Cas}}}{R_{DD}^4(\phi)} + \Lambda_{\text{IR}}(t) \mathcal{F}_{\text{GW}} \left(\frac{v_{\text{br}}}{M_5^{3/2}}, m_{\text{GW}} R_{DD}(\phi) \right) + \frac{\xi}{M_5^2} K_{\text{Weyl}}(r_b, t) \langle \Phi^2 \rangle(\phi). \quad (68)$$

The explicit time dependence at fixed ϕ enters through two channels: $\Lambda_{\text{IR}}(t) \propto \sqrt{K_{\text{Weyl}}} \propto R_h^{-2}$ in the Goldberger–Wise prefactor and $K_{\text{Weyl}}(r_b, t) \propto R_h^{-4}$ in the curvature term. These two channels are built from the same curvature invariant but carry *different* powers of R_h (one the square root of the other), so they are not proportional and scale independently with the accretion history. In the quasi-static regime $R_h \sim 1/(\beta H_0) \gg H_0^{-1}$ the curvature term $V_{\text{curv}} \propto R_h^{-4}$ is parametrically smaller than the Goldberger–Wise term $V_{\text{GW}} \propto R_h^{-2}$, so the latter controls the location of the minimum and the curvature term enters only as a subleading correction. The Casimir piece is time-independent at fixed ϕ . A purely negative graviton Casimir contribution is destabilizing; stabilization of the minimum is provided by the *net* positive quartic vacuum term $\mathcal{A}/R_{DD}^4$ in the toy potential of Section 4.4 (which may include the Casimir piece together with fermion or counterterm contributions). A consistency check that the assumed minimum (72) survives once both curvature channels are included is left to the numerical treatment deferred to future work.

4.2 Kretschmann Invariant as Dynamical IR Cutoff

Substituting (10) at $r = r_b$ and using the tracking solution (29) with $R_h^2 = \mu$ (slow-AdS limit),

$$K_{\text{Weyl}}(r_b, t) = \frac{72\mu(t)^2}{r_b^8} = \frac{72}{c^8 R_h(t)^4}. \quad (69)$$

The infrared cutoff that enters V_{GW} must have dimensions $[\text{mass}]^4$ (an energy density). The natural combination of bulk geometric inputs and the only available mass scale (M_5) that achieves these dimensions and depends non-trivially on R_h is

$$\Lambda_{\text{IR}}(t) \equiv M_5^2 \sqrt{K_{\text{Weyl}}(r_b, t)} = \sqrt{72} M_5^2 c^{-4} R_h(t)^{-2}. \quad (70)$$

Dimensional analysis. M_5 has $[\text{mass}]^1$ in 5D units; $\sqrt{K_{\text{Weyl}}}$ has $[\text{mass}]^2$. The combination $M_5^2 \sqrt{K_{\text{Weyl}}}$ thus has $[\text{mass}]^4$ dimensions of an energy density. Throughout, M_5 denotes the *fixed* fundamental 5D Planck mass, treated as a field-independent constant. In the 6D completion of Appendix D it is identified with the constant combination built from M_6 (equivalently $M_5^3 \equiv 2\pi M_6^4 \ell_r$ with ℓ_r static), and *not* with the radion-dependent effective coupling $M_{5,\text{eff}}^3 \propto R_{DD}$ that arises on reducing over the dark circle. This distinction is essential: substituting the field-dependent $M_{5,\text{eff}}$ into (70) would change the scaling derived below from $R_{DD} \propto c R_h^{1/2}$ to $R_{DD} \propto c^{6/7} R_h^{3/7}$ (see Appendix D). **A minimal choice under the minimality assumption.** Among combinations $M_5^a K_{\text{Weyl}}^b$ with a, b rational and $a + 4b = 4$ (since K_{Weyl} has dimension $[\text{mass}]^4$), a minimal choice consistent with curvature-squared operators and the stated minimality assumption (AdS scale k not included in the cutoff) selects $b = 1/2$, $a = 2$, i.e., (70). In particular, the dimensionally simplest alternative $b = 1$, $a = 0$ (i.e. $\Lambda_{\text{IR}} \propto K_{\text{Weyl}} \propto R_h^{-4}$) would yield $n = 1$ rather than $n = 1/2$; the exponent is therefore selected by, not independent of, the choice $b = 1/2$. Other combinations are possible but would change the parametric scaling of Λ_{IR} . (Note that the AdS curvature scale k is not included in this construction; combinations involving k are possible but would alter the scaling.) The scaling

$$\Lambda_{\text{IR}}(t) \propto \sqrt{K_{\text{Weyl}}(r_b, t)} \propto c(t)^{-4} R_h(t)^{-2} \quad (71)$$

therefore follows from this minimal choice under the additional assumption that the infrared scale is obtained from the square root of the leading curvature invariant of the bulk Weyl tensor.

4.3 Dynamical Minimum and the Origin of the $n = 1/2$ Scaling

The dark-dimension swampland relation (2) reads, in the conventions of (68) (with Λ_{IR} a $[\text{mass}]^4$ energy-density scale),

$$R_{DD,\text{min}} \sim \Lambda_{\text{IR}}^{-1/4}. \quad (72)$$

This is dimensionally consistent: $R_{DD,\text{min}}$ has $[\text{length}] = [\text{mass}]^{-1}$, while $\Lambda_{\text{IR}}^{-1/4}$ has $[\text{mass}]^{-1}$. The relation (72) is taken as the dark-dimension scaling input of Montero–Vafa–Valenzuela (8), fixing the *location* of the compactification scale in terms of the infrared cutoff. We do not derive it from a microscopic compactification; Section 4.4 instead exhibits a phenomenological toy potential whose extremum *realizes* this scaling, subject to a positivity condition

on the net quartic vacuum term. The potential (68) enters the dynamics through the radion mass (83) governing the adiabatic tracking of this slowly moving minimum, which we assume to be positive ($m_{\text{rad}}^2 > 0$). Combining (71) with (72) immediately gives

$$R_{DD,\text{min}}(t) \propto c(t) R_h(t)^{1/2}. \quad (73)$$

Arithmetic decomposition of $n = 1/2$. The exponent $n = 1/2$ is the product of two factors:

$$n = (-1/4) \times (-2) = \frac{1}{2} \quad (\text{swampland exponent} \times \text{curvature scaling}). \quad (74)$$

The first factor is fixed by the Montero–Vafa–Valenzuela construction (2). The second factor is fixed by the scaling of $\Lambda_{\text{IR}} \propto R_h^{-2}$ from the curvature invariant. Neither is adjustable (modulo the minimality assumption for Λ_{IR}).

Why Λ_{IR} scales as R_h^{-2} . The combination $\sqrt{K_{\text{Weyl}}} \propto R_h^{-2}$ arises because $K_{\text{Weyl}} \propto \mu^2/r^8$, and along the tracking $r_b \propto R_h$ with $R_h^2 = \mu$, so

$$K_{\text{Weyl}} \propto \frac{\mu^2}{r_b^8} = \frac{R_h^4}{c^8 R_h^8} = \frac{1}{c^8 R_h^4}. \quad (75)$$

The square root in (70) reduces the R_h^{-4} scaling to R_h^{-2} , which is precisely what is needed to combine with the $-1/4$ swampland exponent to produce $n = +1/2$.

Limiting cases. In the strictly stationary limit $\partial_t \mu = 0$ (constant black-hole mass), R_h is constant. However, a truly static compactification radius also requires $\dot{c} = 0$, which occurs only when the brane matter density ρ_{br} is itself constant (see the tracking relation (40)). When both conditions are satisfied, (73) recovers a fixed compactification radius, reproducing the static Goldberger–Wise stabilization.

In the formal limit $\mu \rightarrow 0$ the tracking ansatz $r_b = cR_h$ drives the brane radius to zero, so the *on-tracking* curvature $K_{\text{Weyl}} = 72/(c^8 R_h^4)$ diverges and $\Lambda_{\text{IR}} \rightarrow \infty$, forcing $R_{DD,\text{min}} \rightarrow 0$. This is an artifact of the ansatz sending $r_b \rightarrow 0$, not intrinsic curvature growth: at *fixed* radius $K_{\text{Weyl}} = 72\mu^2/r^8 \rightarrow 0$ as $\mu \rightarrow 0$. The correct statement is that the tracking ansatz, and with it the curvature-set cutoff, breaks down as $\mu \rightarrow 0$; a pure empty bulk lies outside the regime of validity.

4.4 An Explicit Potential Realizing the Scaling Minimum

The scaling (72) can be realized, rather than merely imposed, by a phenomenological toy potential of the form (68) whose extremum reproduces it. This is a consistency demonstration, not a microscopic derivation: the coefficients below are not computed from a fully specified Goldberger–Wise/Casimir sector. In the quasi-static regime the curvature term $V_{\text{curv}} \propto K_{\text{Weyl}} \propto \Lambda_{\text{IR}}^2$ (up to inverse powers of M_5 and the dimensionally reduced $\langle \Phi^2 \rangle$ in the prefactor) lies parametrically below $V_{\text{GW}} \propto \Lambda_{\text{IR}}$, so the minimum is controlled by the two leading contributions. Writing $R \equiv R_{DD}(\phi)$ and taking the Goldberger–Wise profile in its logarithmic form (the standard $m_{\text{GW}}R \ll 1$ limit recalled in Section 4),

$$V_{\text{eff}}(R) = \frac{\mathcal{A}}{R^4} + \Lambda_{\text{IR}} [\alpha \ln(m_{\text{GW}}R) + \beta_{\text{GW}}], \quad (76)$$

where $\mathcal{A}$ is the *net* coefficient of the quartic vacuum term and $\alpha, \beta_{\text{GW}}$ are dimensionless (β_{GW} is not to be confused with the accretion parameter β). The logarithm is not

optional: balancing the Λ_{IR} -weighted term against the Λ_{IR} -independent quartic term forces $\mathcal{F}'_{\text{GW}} \propto R^{-1}$ if the extremum is to scale as $\Lambda_{\text{IR}}^{-1/4}$.

Extremizing,

$$V'_{\text{eff}}(R) = -\frac{4\mathcal{A}}{R^5} + \frac{\alpha\Lambda_{\text{IR}}}{R} = 0 \quad \implies \quad R_{DD,\text{min}} = \left(\frac{4\mathcal{A}}{\alpha}\right)^{1/4} \Lambda_{\text{IR}}^{-1/4}, \quad (77)$$

which *realizes* the dark-dimension scaling (72) as an extremum condition rather than an external input. Using $\alpha\Lambda_{\text{IR}} = 4\mathcal{A}R_{DD,\text{min}}^{-4}$,

$$V''_{\text{eff}}(R_{DD,\text{min}}) = \frac{16\mathcal{A}}{R_{DD,\text{min}}^6}, \quad m_{\text{rad}}^2 = \frac{R_{DD,\text{min}}^2}{f_\phi^2} V''_{\text{eff}} = \frac{4\alpha\Lambda_{\text{IR}}}{f_\phi^2}, \quad (78)$$

so the extremum is a genuine minimum with $m_{\text{rad}}^2 > 0$ *provided* $\mathcal{A} > 0$ (and hence $\alpha > 0$). The minimum value is $V_{\text{min}} = \Lambda_{\text{IR}}[\frac{\alpha}{4} + \alpha \ln(m_{\text{GW}} R_{DD,\text{min}}) + \beta_{\text{GW}}] = O(\Lambda_{\text{IR}})$, positive for suitable β_{GW} : a positive-energy de Sitter-like vacuum, exactly the configuration whose compatibility with the refined de Sitter conjecture is discussed in Section 6. The neglected curvature term is parametrically subleading in the quasi-static regime: using $K_{\text{Weyl}} = \Lambda_{\text{IR}}^2/M_5^4$ one has $V_{\text{curv}} = \xi M_5^{-6} \langle \Phi^2 \rangle \Lambda_{\text{IR}}^2$, so—up to the inverse powers of the fundamental mass scale M_5 and the dimensionally reduced $\langle \Phi^2 \rangle$ carried by the prefactor—it scales as Λ_{IR}^2 while $V_{\text{GW}} \propto \Lambda_{\text{IR}}$, and is therefore suppressed by one power of Λ_{IR} relative to the leading balance. The resulting fractional shift of $R_{DD,\text{min}}$ is therefore of order Λ_{IR} times a model-dependent coefficient built from ξ , $\langle \Phi^2 \rangle(\phi)$, and the fundamental mass scale; we do not evaluate it here, and writing it as $O(\Lambda_{\text{IR}}/m_{\text{GW}}^4)$ presumes that this coefficient combination reduces to $\sim m_{\text{GW}}^{-4}$, which is not established. The expectation that the Goldberger–Wise term controls the location of the minimum—and an explicit check that the assumed minimum survives once the curvature term is restored—is left to the numerical treatment deferred to future work.

Two features of (76) are worth emphasizing. First, stabilization requires the net quartic coefficient to be *positive*, $\mathcal{A} > 0$; a purely negative graviton Casimir contribution ($\mathcal{A} < 0$) yields a maximum, so a net-positive vacuum contribution (e.g. a fermion-dominated or SUSY-broken tower, or a brane counterterm) is needed—consistently with the positive dark energy of the dark-dimension scenario. Second, with the logarithmic profile the quartic term is the stabilizing contribution and the Goldberger–Wise term the destabilizing one, the reverse of a naive R_{DD}^{-4} -Casimir-versus-quadratic-Goldberger–Wise expectation. A derivation of $\mathcal{A}$ and α from the microscopic Casimir and Goldberger–Wise boundary data, and inclusion of the full curvature term, remain for future work.

5 Adiabatic Evolution and Stability Analysis

5.1 Adiabatic Theorem for the Radion

The minimum (73) drifts with rate

$$\frac{\dot{R}_{DD,\text{min}}}{R_{DD,\text{min}}} = \frac{\dot{c}}{c} + \frac{1}{4} \dot{t} \beta H_0, \quad (79)$$

with $\dot{t}$ the on-tracking redshift factor of Section 3.3, $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}} = O(1)$ (the slow-motion form $c/\sqrt{c^2-1}$ holds only in the sublimit $\dot{r}_b^2 \ll f$). The adiabatic condition is

therefore

$$\eta_{\text{ad}} \sim \frac{|\dot{c}/c + \frac{1}{4}\dot{t}\beta H_0|}{m_{\text{rad}}} \ll 1. \quad (80)$$

If the stabilized radion is heavy compared with the cosmological scale, $m_{\text{rad}} \gg H_0$, then since $\beta \ll 1$ the adiabatic condition is comfortably satisfied provided $\dot{t}$ is finite—that is, away from the window endpoints $c \rightarrow 1^+$ and $n_\rho \rightarrow 1^-$, where $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}}$ diverges (equivalently, the operative requirement is $m_{\text{rad}} \gg \dot{t}\beta H_0$ with $\dot{t}$ finite): the minimum moves only on the slow accretion timescale set by $\dot{c}/c + \frac{1}{4}\dot{t}\beta H_0 = O(\dot{t}\beta H_0)$ (cf. (54)). The radion therefore tracks the slowly moving minimum with negligible lag.

The radion mass-squared at the instantaneous minimum follows from the toy potential of Section 4.4,

$$V_{\text{eff}}(R) = \frac{\mathcal{A}}{R^4} + \Lambda_{\text{IR}}[\alpha \ln(m_{\text{GW}}R) + \beta_{\text{GW}}], \quad (81)$$

whose extremum satisfies

$$\alpha \Lambda_{\text{IR}} = \frac{4\mathcal{A}}{R_{DD,\text{min}}^4}, \quad (82)$$

and therefore

$$V_{\text{eff}}''(R_{DD,\text{min}}) = \frac{16\mathcal{A}}{R_{DD,\text{min}}^6}, \quad m_{\text{rad}}^2 = \frac{R_{DD,\text{min}}^2}{f_\phi^2} V_{\text{eff}}'' = \frac{4\alpha \Lambda_{\text{IR}}}{f_\phi^2} + O(\xi). \quad (83)$$

Thus the toy extremum is stable when $\mathcal{A} > 0$ and $\alpha > 0$. In this realization the net quartic vacuum term supplies the stabilizing curvature of the potential, while the logarithmic Goldberger–Wise term fixes the scaling of the minimum; a purely negative graviton Casimir term would not stabilize the radius by itself.

The equation of motion for the canonically normalized radion on a Friedmann–Robertson–Walker background reads

$$\ddot{\phi} + 3H_b \dot{\phi} + \partial_\phi V_{\text{eff}}(\phi; t) = 0, \quad (84)$$

with H_b the brane Hubble rate from (25).

5.2 Linear Stability of the Coupled Brane–Radion System

Perturb the combined system around the attractor $\{r_b(t) = c(t)R_h(t), \phi(t) = \phi_{\text{min}}(t)\}$ by writing $r_b(t) = c(t)R_h(t)(1 + \varepsilon(t))$ and $\phi(t) = \phi_{\text{min}}(t) + \delta\phi(t)$, with $|\varepsilon| \ll 1$ and $|\delta\phi| \ll 1$.

The Israel junction condition (25) is a first-order constraint that reduces the brane sector to a single physical degree of freedom. Linearizing this constraint, including the matter-density perturbation

$$\frac{\delta\rho_{\text{br}}}{\rho_{\text{br}}} = -n_\rho \varepsilon,$$

and imposing the high-energy force-balance relation (40), gives the constrained decay equation

$$\dot{\varepsilon} = -H_b|_0 \varepsilon + O(\beta H_0 \varepsilon). \quad (85)$$

The background Hubble rate on the tracking solution is

$$H_b^2|_0 = \frac{1}{c^4 R_h^2} \left(\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right). \quad (86)$$

Thus the brane perturbation decays whenever the tracking background is expanding. For $B > 0$, this requires

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}. \quad (87)$$

The familiar window $1 < c < \sqrt{2}$ is recovered only in the special vacuum-like case $n_\rho = 0$.

The radion sector remains second-order. The curvature coupling $\frac{\xi}{M_5^2} K_{\text{Weyl}} \Phi^2$ (see Section 4.1) generates off-diagonal mixing terms at leading order in the perturbation ε (the relative variation $\delta K_{\text{Weyl}}/K_{\text{Weyl}} \sim -8\varepsilon$). Under the leading-order decoupling approximation (valid when the curvature coupling is not parametrically large), the joint system is described by a 3×3 stability matrix (see Appendix B for the reduced matrix and eigenvalues). Because the brane sector is reduced to a single mode fixed by the first-order Israel constraint, the linearized brane equation receives no backreaction from $\delta\phi$ and the stability matrix is triangular; what we establish is therefore the *separate* stability of the constrained brane mode and the radion mode, not a fully coupled result. A complete coupled analysis would require restoring radion backreaction on the brane stress tensor and the junction conditions, which we leave to future work.

Within the zeroth-order quasi-static parameter regime $\dot{R}_h > 0$, $n_\rho < 1$, and

$$1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho},$$

the linearized system is stable at leading order, provided the radion mass satisfies $m_{\text{rad}}^2 > 0$ and the curvature-induced brane–radion mixing remains perturbative. The old window $1 < c < \sqrt{2}$ applies only to the special case $n_\rho = 0$.

The boundary $n_\rho = 1$, the static limit $\dot{R}_h \rightarrow 0$, and the region outside the generalized window (87) require separate analysis. The point $n_\rho = 2$ lies outside the high-energy branch $n_\rho < 1$ and is only a singular point of the zeroth-order drift expression, not a boundary of the tracking window.

5.3 Effective 4D Description

In the present effective treatment, the 5D Schwarzschild–Tangherlini geometry supplies the curvature scale entering the radion potential, while the compact dark dimension is represented in the reduced low-energy Kaluza–Klein ansatz by the modulus $R_{DD}(x)$. A fully geometric model containing both a noncompact black-hole radial bulk direction and an independent compact dark dimension would require a higher-dimensional (likely 6D) completion. We leave that embedding to future work. Accordingly, the Kaluza–Klein relation $M_{\text{Pl}}^2 = M_5^3 R_{DD}$ used below is *not* derived from the black-hole bulk of Section 2, whose fifth dimension is the noncompact radial direction r ; it belongs to a logically separate, phenomenological reduction sector whose fifth dimension is the compact circle y of size R_{DD} . Identifying the scale in that relation with the bulk M_5 is an assumption of the present effective treatment, to be justified only by an explicit higher-dimensional (likely 6D) completion. A candidate 6D completion that reduces this relation to a small set of explicitly stated conditions—finiteness and static behaviour of the radial zero-mode measure ℓ_r , and a wrapped-brane junction consistent with the 5D one—is set up in Appendix D; within that completion $M_{\text{Pl}}^2 \propto R_{DD}$ follows once those conditions are verified, but it is not derived from the 5D bulk of Section 2. At leading order in the slow-accretion and thin-brane approximations we assume warp-factor corrections from the radial geometry decouple from the dark-dimension modulus.

For the Kaluza–Klein ansatz

$$ds_{(5)}^2 = g_{\mu\nu}(x)dx^\mu dx^\nu + R_{DD}^2(x)dy^2, \quad y \in [0, 1), \quad (88)$$

the dimensional reduction of the Ricci scalar is

$$R^{(5)} = R^{(4)} - \frac{2}{R_{DD}} \square^{(4)} R_{DD}. \quad (89)$$

Substituting into the Einstein–Hilbert piece of (3) and integrating over the compact direction gives

$$S_{\text{EH}}^{(5)} = \frac{M_5^3}{2} \int d^4x \sqrt{-g^{(4)}} R_{DD}(x) \left(R^{(4)} - \frac{2}{R_{DD}} \square^{(4)} R_{DD} \right). \quad (90)$$

The $\square^{(4)} R_{DD}$ term is a total covariant divergence and integrates to a boundary contribution. The reduced action is the Jordan-frame Einstein–Hilbert form

$$S_{\text{EH}}^{(4)} = \frac{M_5^3}{2} \int d^4x \sqrt{-g^{(4)}} R_{DD}(x) R^{(4)}, \quad (91)$$

whose coefficient defines a time-dependent effective Planck mass

$$M_{\text{Pl}}^2(t) \equiv M_5^3 R_{DD}(t), \quad G_4(t) = \frac{1}{8\pi M_{\text{Pl}}^2(t)} = \frac{1}{8\pi M_5^3 R_{DD}(t)}. \quad (92)$$

We stress that the dimensional reduction (89)–(91) leading to (92) integrates over the compact circle y of size R_{DD} , which is *not* the radial direction r of the black-hole bulk of Section 2. In the strictly 5D setup there is only one extra dimension (the noncompact r), and it cannot simultaneously play the role of the compact y in (91). The relation (92), and therefore the central phenomenological result $G_4(t) \propto 1/R_{DD}(t)$ together with the drift bounds of Section 6, are valid only in a two-extra-dimension (e.g. 6D) completion in which the black-hole radial direction and the compact dark dimension coexist and the identification of the reduction scale with M_5 is justified. Within the present 5D treatment they should be read as the anticipated form of the result in that completion rather than as a consequence derived from the bulk of Section 2.

Since $R_{DD} \propto c(t)R_h^{1/2}$ and $R_h \propto \mu^{1/2}$, and using $\dot{R}_h/R_h = \dot{t} \frac{1}{2} \beta H_0$ in proper time, we obtain

$$\frac{\dot{R}_{DD}}{R_{DD}} = \frac{\dot{c}}{c} + \frac{1}{4} \dot{t} \beta H_0, \quad (93)$$

leading to the 4D gravitational coupling drift

$$\frac{\dot{G}_4}{G_4} = -\frac{\dot{c}}{c} - \frac{1}{4} \dot{t} \beta H_0, \quad (94)$$

with $\dot{t}$ the on-tracking redshift factor of Section 3.3, $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}} = O(1)$ (the slow-motion form $c/\sqrt{c^2-1}$ holds only in the sublimit $\dot{r}_b^2 \ll f$).

Effective Einstein equations. Varying the reduced action with respect to $g_{\mu\nu}$ produces the Jordan-frame field equations

$$R_{DD} G_{\mu\nu}^{(4)} = \frac{1}{M_5^3} T_{\mu\nu}^m + \nabla_\mu \nabla_\nu R_{DD} - g_{\mu\nu} \square^{(4)} R_{DD}. \quad (95)$$

Along the tracking solution the Jordan-frame derivative terms are controlled by the full drift rate of R_{DD} . Since $\dot{c}/c = O(\dot{t}\beta H_0)$ by (54), the derivative terms scale as $O(\dot{t}^2\beta^2 H_0^2)$ away from the near-horizon boundary $c \rightarrow 1$. They therefore reduce to small Brans–Dicke-like corrections within the quasi-static regime.

Controlled decoupling and local screening. The cosmological drift rate is

$$\frac{\dot{G}_4}{G_4} = -\frac{\dot{c}}{c} - \frac{1}{4}\dot{t}\beta H_0. \quad (96)$$

Both contributions to the cosmological drift are $O(\dot{t}\beta H_0)$ on the tracking solution (cf. (54)), so the total $|\dot{G}_4/G_4|$ is itself $O(\dot{t}\beta H_0)$. Local bounds on $\dot{G}/G$ constrain this drift directly only in the unscreened case. If the radion is locally screened, the locally measured $G_{4,\text{loc}}$ can be decoupled from the cosmological value, and local tests no longer bound β without specifying the residual screening leakage.

Note that $c(t)$ is the global embedding ratio $r_b(t)/R_h(t)$. Locally, the radion can additionally be screened by a chameleon-like coupling

$$V_{\text{eff}}^{\text{cham}}(\phi; \rho_m) = V_{\text{eff}}(\phi; M(t)) + \beta_\phi \frac{\rho_m}{M_{\text{Pl}}} \phi, \quad (97)$$

under which the local radion mass is enhanced with density,

$$m_{\text{rad,loc}}^2(\rho_m) \sim m_{\text{rad}}^2 \left(\frac{\rho_m}{\rho_{\text{cos}}} \right)^p, \quad p > 0. \quad (98)$$

For $\rho_{\text{lab}}/\rho_{\text{cos}} \sim 10^{30}$ and $p \sim 1$, the local mass-squared is enhanced by $\sim 10^{30p}$ (the mass itself by $\sim 10^{15p}$), exponentially suppressing fifth-force fluctuations and locking the *local* value of R_{DD} to a high-density minimum. In this regime the locally measured $G_{4,\text{loc}}$ is effectively constant while the cosmological G_4 (determined by the global $c(t)$) continues to evolve according to (96). In such a screened completion the local bounds on $\dot{G}_{4,\text{loc}}$ could in principle be weakened relative to the cosmological drift. We stress, however, that the density-dependent enhancement (98) is assumed schematically and is *not* derived from the radion potential (68): establishing genuine decoupling requires specifying the radion–matter coupling and verifying that the screening mechanism is compatible with the same potential responsible for cosmological tracking. We therefore treat the screened scenario as a possibility to be substantiated, not as an established feature of the model.

6 Phenomenological Implications and Consistency Checks

6.1 Running of Fundamental Scales

In a higher-dimensional completion in which a compact dark dimension of size R_{DD} coexists with the noncompact radial bulk direction and obeys the Kaluza–Klein relation $M_{\text{Pl}}^2 \propto M_5^3 R_{DD}$, the tracking solution implies a drift of G_4 , estimated below. All phenomenological statements in this section are conditional on that higher-dimensional assumption *and* on the conditions of Appendix D (finite, static ℓ_r and a wrapped-brane junction consistent with the 5D one); moreover, as noted below, the drift *coefficient* $\gamma(n_\rho)$ is only a zeroth-order-tracking estimate. They should therefore be read as conditional order-of-magnitude expectations, not as controlled predictions.

On the tracking solution the drift of the compactification radius receives two contributions of the same order: the induced drift of $c(t)$ and the explicit accretion-driven growth of the horizon. These combine to give

$$\frac{\dot{R}_{DD}}{R_{DD}} = \gamma(n_\rho) \dot{t} \beta H_0, \quad \gamma(n_\rho) \equiv \frac{n_\rho}{4(2 - n_\rho)}, \quad (99)$$

where $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}} = O(1)$ is the on-tracking redshift factor. Consequently the effective 4D Newton constant drifts as

$$\frac{\dot{G}_4}{G_4} = -\gamma(n_\rho) \dot{t} \beta H_0. \quad (100)$$

This expression should be read as a zeroth-order adiabatic estimate only. The controlled $O(\beta)$ Vaidya calculation in Appendix C shows that the accretion forcing, the brane–bulk energy exchange, and the kinematic terms $\ddot{c}$, $\dot{c}\dot{R}_h$, $\ddot{R}_h$ all enter the brane acceleration $\ddot{r}_b/r_b$ at the *same* order, $O(\beta^2 H_0^2)$, as the terms that fix the drift of $c(t)$. They therefore shift the drift coefficient—and hence $\gamma(n_\rho)$ —by an *order-unity relative* amount, not by a relative $O(\beta)$. What survives at this order is only the parametric scaling: every contribution to $\dot{R}_{DD}/R_{DD}$ is $O(\beta H_0)$, so the corrections shift $\dot{c}/c$ by $O(\beta H_0)$ and $\dot{G}_4/G_4$ remains $O(\beta H_0)$ regardless. Consequently the closed form $\gamma(n_\rho) = n_\rho/[4(2 - n_\rho)]$ is a zeroth-order-tracking estimate whose precise value (and even the relative weight of its two contributions) is not fixed until the full first-order tracking problem of Appendix C is solved; only the order of magnitude of the bound below is controlled.

Note that $\gamma(0) = 0$: in the vacuum-like case $n_\rho = 0$ (which recovers the familiar window $1 < c < \sqrt{2}$) the induced drift of $c(t)$ exactly cancels the explicit accretion term at this order, so there is no leading-order drift of G_4 . For $0 < n_\rho < 1$ the drift is nonzero. Its phenomenological consequences depend on whether the radion is locally screened.

Unscreened case. If there is no chameleon-type density-dependent screening, the cosmological drift $\dot{G}_4/G_4$ is the quantity probed by local tests, and identifying it with the lunar-laser-ranging bound ($|\dot{G}/G| \lesssim 10^{-13} \text{ yr}^{-1}$) gives

$$\beta \lesssim \frac{10^{-13} \text{ yr}^{-1}}{\gamma(n_\rho) \dot{t} H_0} \approx \frac{1.4 \times 10^{-3}}{\gamma(n_\rho) \dot{t}}, \quad (101)$$

The case $n_\rho = 2$ is degenerate (cf. (54)) and lies outside the regime in which a positive high-energy tracking root exists. The numerical bounds above are therefore quoted only for the restricted class of matter with $n_\rho < 1$ that supports the adiabatic tracking solution derived in Section 3.3. Ordinary dust-like ($n_\rho = 3$) and radiation-like ($n_\rho = 4$) conserved matter do not admit a consistent positive high-energy tracking root in the present simplified force-balance model and would require additional sources or a controlled departure from strict tracking.

Screened case. If the radion is locally screened along the lines of Section 5 ($m_{\text{rad,loc}}^2 \gg m_{\text{rad}}^2$ in high-density environments), the locally measured $G_{4,\text{loc}}$ is effectively constant while the cosmological G_4 continues to drift at the rate above. Lunar-laser-ranging then does *not* bound β directly; instead the bound comes from genuinely cosmological probes of G_4 – BBN constraints on $\Delta \ln G_4$ between BBN and today (Section 6, below), and CMB constraints on a varying gravitational coupling at recombination. These are typically order 10^{-2} at the integrated level, translating to weaker (order-of-magnitude) constraints on

the time-averaged $\bar{\beta}$. The two regimes therefore yield qualitatively different phenomenology: in the unscreened case β is bounded near 10^{-3} ; in the screened case β may sit closer to 10^{-2} , with the BBN/CMB integrated bounds replacing local LLR.

Big Bang Nucleosynthesis constrains the *integrated* change in G_4 . Because $G_4 \propto c^{-1} R_h^{-1/2}$, we have

$$\Delta \ln G_4 = -\Delta \ln c - \frac{1}{2} \Delta \ln R_h. \quad (102)$$

A bound on the time-averaged accretion parameter $\bar{\beta}$ that is derived from an assumed fractional change in R_h alone therefore requires an assumption about the integrated drift of $c(t)$ (either that it is screened, negligible, or independently constrained). Future improvements in BBN and CMB-based determinations of G_4 will tighten these bounds once the separate contributions from Δc and ΔR_h can be disentangled. The next decade of clock-network experiments is projected to improve the local bound on $|\dot{G}/G|$ by one to two orders of magnitude.

6.2 Consistency with Swampland Conjectures

The distance conjecture and dark-dimension scaling are compatible with the construction. The canonical field excursion of the radion between an early time t_i and the present is

$$\Delta\phi(t_i \rightarrow \text{now}) = f_\phi \ln \frac{R_{DD,\text{now}}}{R_{DD,i}} = f_\phi \ln \frac{c_{\text{now}}}{c_i} + \frac{f_\phi}{2} \ln \frac{R_{h,\text{now}}}{R_{h,i}}, \quad (103)$$

Compatibility with the distance conjecture therefore requires

$$\left| \ln \frac{c_{\text{now}}}{c_i} + \frac{1}{2} \ln \frac{R_{h,\text{now}}}{R_{h,i}} \right| \lesssim \frac{2M_{\text{Pl}}}{f_\phi}. \quad (104)$$

This is not automatic; it imposes a bound on the integrated accretion history and on the allowed drift of $c(t)$. Compatibility with the distance conjecture is therefore conditional on the global cosmological evolution of the bulk mass parameter.

The refined de Sitter swampland conjecture, however, is in tension with a stable positive-energy minimum of the radion potential. The conjecture requires that for any positive-energy de Sitter vacuum,

$$\frac{|V_{,\phi}|}{V} \geq c_{\text{dS}} \frac{1}{M_{\text{Pl}}} \quad \text{or} \quad -\frac{V_{,\phi\phi}}{V} \geq c'_{\text{dS}} \frac{1}{M_{\text{Pl}}^2}. \quad (105)$$

Along the tracking trajectory the first derivative vanishes by construction, $V_{,\phi}|_{\phi_{\text{min}}} = 0$. At a stable minimum we also have $V_{,\phi\phi} > 0$. For a positive vacuum energy $V > 0$ this immediately implies

$$-\frac{V_{,\phi\phi}}{V} < 0,$$

which violates the second refined de Sitter condition.

The model therefore uses the dark-dimension / distance-conjecture scaling $R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}$ as motivation for the dynamical compactification radius, but a stable positive-energy radion minimum generically conflicts with the refined de Sitter swampland conjecture. This tension can be evaded if the solution is interpreted as time-dependent (the minimum is slowly rolling), metastable (a long-lived false vacuum), or not a true de Sitter

vacuum (the effective cosmological constant is only approximately constant on cosmological timescales). Compatibility with the refined de Sitter conjecture is therefore left as an open question that depends on the detailed global structure of the potential and the interpretation of the late-time cosmology.

The distance conjecture bound on the field excursion is satisfied whenever (104) holds.

6.3 Testable Signatures

Subject to the scope limitations discussed above, the construction suggests the following signatures:

1. A slow drift of G_4 tied to bulk accretion (contingent on the Kaluza–Klein reduction of Section 5, i.e. on the higher-dimensional completion in which $G_4(t) \propto 1/R_{DD}(t)$ holds, and not on the 5D bulk of Section 2; see the caveat there). A quantitative prediction requires the explicit $\beta(t)$, $c(t)$, and equation-of-state history. Parametrically,

$$\Delta \ln G_4 \simeq - \int \gamma(n_\rho) \dot{t} \beta H_0 d\tau, \quad (106)$$

so any induced change in the BBN helium abundance scales as

$$\frac{\Delta Y_p}{Y_p} \sim 0.1 \Delta \ln G_4, \quad (107)$$

up to the usual BBN response coefficients. In the vacuum-like case $n_\rho = 0$, $\gamma = 0$ and the leading-order contribution vanishes.

2. A KK graviton mass spacing $m_{KK} \sim 1/R_{DD}(t) \propto c(t)^{-1} R_h(t)^{-1/2}$ that evolves on the accretion timescale. At present, $R_{DD} \sim 1\text{--}10 \mu\text{m}$ gives $m_{KK} \sim 0.02\text{--}0.2 \text{ eV}$. Future sub-millimeter gravity experiments may be sensitive to the drift

$$\frac{\dot{m}_{KK}}{m_{KK}} = -\frac{\dot{c}}{c} - \frac{1}{4} \dot{t} \beta H_0. \quad (108)$$

3. Speculatively, and outside the quasi-static regime established here, a stochastic gravitational-wave background from radion oscillations during early-universe transitions in $\mu(t)$, with characteristic frequency in the mHz LISA band for transitions at electroweak temperature. This lies in the $\beta \ll 1$ -relaxed regime listed as future work in Section 7 and is not derived in the present treatment.

7 Discussion and Outlook

We have presented a quasi-static mechanism for modulus tracking in a 5D braneworld toy model sourced by a slowly accreting bulk black hole. The central assumptions are: (i) slow accretion ($\beta \ll 1$), treated at leading adiabatic order, (ii) $\mathbb{Z}_2$ -symmetric brane embedding with the brane equation of motion derived at leading order from the Israel junction conditions (with the controlled $O(\beta)$ Vaidya calculation given in Appendix C), (iii) Goldberger–Wise stabilization with brane-localized potentials, and (iv) the dark-dimension swampland scaling $R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}$.

The construction has three notable features. First, it combines an adiabatically time-dependent bulk geometry, a moving brane governed (at leading adiabatic order) by the

Israel junction conditions, and a stabilization potential whose infrared cutoff is supplied by the local bulk Weyl curvature. Second, the brane position adiabatically tracks the parent horizon via $r_b(t) = c(t)R_h(t)$, so that the dark-radiation and spatial-curvature terms scale together as $R_h^{-2}(t)$ up to the slow drift of $c(t)$, generating a slowly drifting compactification radius $R_{DD,\min}(t) \propto c(t)R_h(t)^{1/2}$. Third, the exponent $n = 1/2$ follows from the curvature-driven cutoff $\Lambda_{\text{IR}} = M_5^2 \sqrt{K_{\text{Weyl}}} \propto R_h^{-2}$ combined with the swampland exponent $-1/4$, but is contingent on the minimality assumption used to construct Λ_{IR} : the alternative $\Lambda_{\text{IR}} \propto K_{\text{Weyl}}$ would give $n = 1$.

Key limitations include the restriction to slow accretion, the leading-order analytic treatment of perturbations, and the absence of full nonlinear numerical evolution. Equally important, the *location* of the radion minimum is motivated by the dark-dimension scaling $R_{DD,\min} \sim \Lambda_{\text{IR}}^{-1/4}$. Section 4.4 exhibits a phenomenological toy potential whose extremum realizes this scaling, but this does not yet constitute a microscopic derivation from a fully specified Goldberger–Wise/Casimir sector. Deriving the required coefficients and signs from a complete higher-dimensional compactification remains future work. Most importantly, self-consistency of the induced Hubble rate (Eq. (56)) pins the bulk horizon to a super-Hubble scale and forces $H_b \sim \dot{t} \beta H_0 \ll H_0$ (with $\dot{t} = \frac{c}{(c^2-1)\sqrt{1-n_\rho}} = O(1)$ on the tracking solution; see Section 3.3), and the weak-AdS assumption combined with this pinning imposes the nonstandard hierarchy $k \ll \beta H_0$ (61): the construction is a consistent quasi-static stabilization mechanism but does not simultaneously describe the observed cosmological expansion. Reconciling the two is the central open problem. Future directions include: (i) numerical integration of the coupled first-order brane–radion system; (ii) generalization to higher-dimensional bulks; (iii) early-universe extensions in which $\beta \ll 1$ may be relaxed. The framework opens the possibility of invariant relations between evolving scales (G_4 , R_{DD} , R_h) that may be testable through cosmological observations.

A Detailed Junction Calculations

A.1 Geometric Preliminaries

For the time-dependent embedding $r_b = r_b(\tau)$ in the bulk metric (8), the timelike four-velocity and outward unit normal are given by (17)–(18). The induced metric on the brane is $h_{\mu\nu} = g_{\mu\nu} - n_\mu n_\nu$, which on the brane takes the FRW form $h_{ab}dx^a dx^b = -d\tau^2 + r_b(\tau)^2 d\Omega_3^2$. The orthonormalization conditions $u^a u_a = -1$, $n^a n_a = +1$, $u^a n_a = 0$ may be verified directly from the explicit components.

A.2 Extrinsic Curvature Components

The non-zero Christoffel symbols of (8) that contribute to the extrinsic curvature (in the static- f limit) are

$$\Gamma_{tr}^t = \frac{f'}{2f}, \quad \Gamma_{tt}^r = \frac{f f'}{2}, \quad \Gamma_{rr}^r = -\frac{f'}{2f}, \quad \Gamma_{\theta_i \theta_i}^r = -r f \hat{g}_{\theta_i \theta_i}, \quad \Gamma_{r \theta_i}^{\theta_i} = \frac{1}{r}, \quad (109)$$

where $f' \equiv \partial_r f = 2\mu/r^3 + 2k^2 r$. Computing $K_{ab} = h_a^c h_b^d \nabla_c n_d$ component by component yields (20) and (21). The angular result follows directly from $K_j^i = (n^r/r) \delta_j^i =$

$(\sqrt{f + \dot{r}_b^2}/r_b) \delta_j^i$, while the temporal component is obtained by computing $u^a u^b \nabla_a n_b$ explicitly and simplifying with the constraint $-f\dot{t}^2 + \dot{r}_b^2/f = -1$.

At zeroth adiabatic order this is consistent, since $\partial_t f = -\dot{\mu}/r^2 = O(\beta)$ is dropped along with the explicit accretion force in (113). A uniformly consistent $O(\beta)$ treatment requires the ingoing-Vaidya form of the bulk metric, including the off-diagonal G^t_r source; this controlled calculation is carried out in Appendix C.

A.3 Israel Conditions and the Modified Friedmann Equation

Imposing (19) with the surface stress-energy tensor

$$S^\mu{}_\nu = -\sigma \delta^\mu{}_\nu + (T^m)^\mu{}_\nu, \quad (T^m)^\mu{}_\nu = \text{diag}(-\rho_{\text{br}}, p_{\text{br}}, p_{\text{br}}, p_{\text{br}}), \quad (110)$$

and applying $\mathbb{Z}_2$ symmetry yields the angular condition (23). Squaring and using $H_b^2 = \dot{r}_b^2/r_b^2$ together with $f = 1 - \mu/r_b^2 + k^2 r_b^2$,

$$H_b^2 + \frac{1}{r_b^2} - \frac{\mu}{r_b^4} + k^2 = \frac{\kappa_5^4}{36}(\sigma^2 + 2\sigma\rho_{\text{br}} + \rho_{\text{br}}^2). \quad (111)$$

The RS tuning $k^2 = \kappa_5^4 \sigma^2/36$ cancels the σ^2 term against the $+k^2$ contribution on the left, leaving

$$H_b^2 = \frac{\kappa_5^4}{36}\rho_{\text{br}}^2 + \frac{\kappa_5^4 \sigma \rho_{\text{br}}}{18} + \frac{\mu}{r_b^4} - \frac{1}{r_b^2}, \quad (112)$$

which is (25).

A.4 Brane Equation of Motion

The second-order brane equation of motion is most cleanly obtained by differentiating the Friedmann constraint (112) with respect to proper time and imposing brane stress-energy conservation $\dot{\rho}_{\text{br}} + 3H_b(\rho_{\text{br}} + p_{\text{br}}) = 0$; by the contracted Gauss–Codazzi identity the result coincides with the $\tau\tau$ -projection of the Israel condition (19). In the high-energy regime $\rho_{\text{br}} \gg \sigma$, writing $\rho_{\text{br}} \propto r_b^{-n_\rho}$ and working at zeroth adiabatic order ($\dot{\mu} = 0$), this gives directly

$$\frac{\ddot{r}_b}{r_b} = (1 - n_\rho) B - \frac{\mu}{r_b^4}, \quad B \equiv \frac{\kappa_5^4 \rho_{\text{br}}^2}{36}, \quad (113)$$

which is Eq. (37). The crucial point is that the matter contribution is differentiated *along* the conserved trajectory (the total r_b -derivative, not the partial derivative at fixed ρ_{br}); this produces the n_ρ -dependent factor, which survives even at $\dot{r}_b = 0$ and fixes the force balance (40). Because the constraint is written in proper time τ , the conversion factor $\dot{t}$ is already incorporated and no separate velocity-dependent friction term appears; the quasi-static condition $\ddot{r}_b = 0$ applied to (113) is the operative tracking condition.

The controlled $O(\beta)$ accretion correction, derived from the ingoing-Vaidya bulk in Appendix C, modifies (113) by an additive forcing

$$\ddot{r}_b = \ddot{r}_b^{\text{static}} - \frac{\mu'(v) \dot{v}^2}{2r_b^2}, \quad \dot{v} \equiv \frac{dv}{d\tau}, \quad (114)$$

of the same order as $\ddot{c}$, $\dot{c} \dot{R}_h$, $\ddot{R}_h$, and the brane–bulk energy-exchange source. The stability of the tracking branch is therefore *not* a friction effect; it follows from the constrained

Friedmann perturbation analysis in Appendix B, where the single physical brane mode has eigenvalue $-H_b|_0$. Equation (113) reduces to the standard high-energy Binétry–Deffayet–Ellwanger–Langlois brane-cosmology result—i.e. the high-energy limit of that equation, in which the linear $\sigma\rho_{\text{br}}$ term has been dropped—in the static limit $\partial_t\mu = 0$.

B Perturbation Matrix and Eigenvalue Analysis

B.1 Constrained Perturbation Matrix

The Israel junction condition (25) is a first-order constraint that reduces the brane sector to a single physical degree of freedom. Linearizing this constraint, including the matter-density perturbation

$$\frac{\delta\rho_{\text{br}}}{\rho_{\text{br}}} = -n_\rho\varepsilon, \quad (115)$$

together with the radion equation about the joint attractor yields an effective 3×3 system in the variables

$$(\varepsilon, \delta\phi, \delta\dot{\phi}),$$

in which the brane velocity $\dot{\varepsilon}$ is algebraically fixed by the first-order constraint.

Working in the high-energy regime $\rho_{\text{br}} \gg \sigma$, the Friedmann constraint (25) reads

$$H_b^2 = B + \frac{\mu}{r_b^4} - \frac{1}{r_b^2}, \quad B \equiv \frac{\kappa_5^4 \rho_{\text{br}}^2}{36}. \quad (116)$$

On the tracking background $r_b = cR_h$, $\mu = R_h^2$, the high-energy force-balance condition (40) gives

$$B = \frac{1}{(1 - n_\rho)c^4 R_h^2}. \quad (117)$$

Therefore the background Friedmann constraint becomes

$$H_b^2|_0 = \frac{1}{c^4 R_h^2} \left(\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right). \quad (118)$$

For $B > 0$, the tracking relation requires

$$n_\rho < 1. \quad (119)$$

In that regime the expanding-background condition is

$$1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}. \quad (120)$$

The familiar $1 < c < \sqrt{2}$ window is recovered only in the special case $n_\rho = 0$.

Now perturb

$$r_b = cR_h(1 + \varepsilon), \quad |\varepsilon| \ll 1, \quad (121)$$

and use $\rho_{\text{br}} \propto r_b^{-n_\rho}$, so that

$$\frac{\delta B}{B} = -2n_\rho\varepsilon. \quad (122)$$

Since

$$H_b = H_b|_0 + \dot{\varepsilon} + O(\varepsilon^2), \quad (123)$$

the left-hand side of the linearized Friedmann constraint gives

$$\delta(H_b^2) = 2H_b|_0 \dot{\varepsilon}. \quad (124)$$

The right-hand side gives

$$\delta \left(B + \frac{\mu}{r_b^4} - \frac{1}{r_b^2} \right) = \left[-2n_\rho B - \frac{4}{c^4 R_h^2} + \frac{2}{c^2 R_h^2} \right] \varepsilon. \quad (125)$$

Substituting (117) yields

$$2H_b|_0 \dot{\varepsilon} = -2 \left(\frac{2-n_\rho}{1-n_\rho} - c^2 \right) \frac{\varepsilon}{c^4 R_h^2}. \quad (126)$$

Using (118), this simplifies to

$$\dot{\varepsilon} = -H_b|_0 \varepsilon. \quad (127)$$

Thus the constrained brane perturbation decays whenever $H_b|_0 > 0$, which is precisely the generalized window (120).

The radion perturbation satisfies, to leading order,

$$\delta\ddot{\phi} + 3H_b|_0 \delta\dot{\phi} + m_{\text{rad}}^2 \delta\phi = \mathcal{S}_{\phi\varepsilon} \varepsilon, \quad (128)$$

where $\mathcal{S}_{\phi\varepsilon}$ encodes curvature-induced mixing from the variation

$$\frac{\delta K_{\text{Weyl}}}{K_{\text{Weyl}}} = -8\varepsilon \quad (129)$$

at fixed R_h . In the leading-order decoupling approximation, valid when the curvature coupling is not parametrically large, the stability matrix takes the schematic form

$$\frac{d}{d\tau} \begin{pmatrix} \varepsilon \\ \delta\phi \\ \delta\dot{\phi} \end{pmatrix} = \begin{pmatrix} -H_b|_0 & 0 & 0 \\ 0 & 0 & 1 \\ \mathcal{S}_{\phi\varepsilon} & -m_{\text{rad}}^2 & -3H_b|_0 \end{pmatrix} \begin{pmatrix} \varepsilon \\ \delta\phi \\ \delta\dot{\phi} \end{pmatrix}. \quad (130)$$

Because the matrix is triangular in the leading decoupling approximation, the eigenvalues are exactly

$$\lambda_{\text{brane}} = -H_b|_0, \quad \lambda_{\text{rad}}^\pm = -\frac{3H_b|_0}{2} \pm \frac{1}{2} \sqrt{9H_b^2|_0 - 4m_{\text{rad}}^2}. \quad (131)$$

Hence the coupled system is linearly stable provided

$$H_b|_0 > 0, \quad m_{\text{rad}}^2 > 0, \quad (132)$$

and the curvature-induced brane–radion mixing is perturbative. The boundary $n_\rho = 1$, the static limit $\dot{R}_h \rightarrow 0$, and the region outside (120) require separate analysis. The point $n_\rho = 2$ lies outside the high-energy branch $n_\rho < 1$ and is only a singular point of the zeroth-order drift expression, not a boundary of the tracking window.

B.2 Decoupling Approximation and Stability of the Radion Sector

The full brane–radion system contains curvature-induced mixing because the radion potential depends explicitly on the Weyl invariant evaluated at the brane. Since

$$K_{\text{Weyl}}(r_b, t) = \frac{72\mu(t)^2}{r_b^8}, \quad (133)$$

a perturbation of the brane position at fixed R_h gives

$$\frac{\delta K_{\text{Weyl}}}{K_{\text{Weyl}}} = -8\varepsilon. \quad (134)$$

This induces off-diagonal entries in the linearized system through the curvature dependence of V_{eff} . Schematically,

$$\delta(\partial_\phi V_{\text{eff}}) = m_{\text{rad}}^2 \delta\phi + \mathcal{M}_{\phi\varepsilon} \varepsilon, \quad (135)$$

where $\mathcal{M}_{\phi\varepsilon}$ is controlled by the curvature coupling ξ , the magnitude of K_{Weyl} , and derivatives of the dimensionally reduced scalar expectation value $\langle\Phi^2\rangle(\phi)$. It is the same mixing coupling as the source $\mathcal{S}_{\phi\varepsilon}$ of (128)–(130), related by $\mathcal{S}_{\phi\varepsilon} = -\mathcal{M}_{\phi\varepsilon}$ (the sign fixed by $\delta\ddot{\phi} = -\delta(\partial_\phi V_{\text{eff}}) + \dots$).

The leading-order decoupling approximation assumes that this mixing is perturbative compared with the diagonal restoring force of the stabilized radion:

$$|\mathcal{M}_{\phi\varepsilon}| \ll m_{\text{rad}}^2 f_\phi. \quad (136)$$

Equivalently, the curvature-induced displacement of the radion minimum caused by a small brane perturbation is required to be much smaller than the intrinsic radion restoring scale. Under this assumption, the brane sector and radion sector may be diagonalized perturbatively.

After imposing the Israel constraint, the brane sector is reduced to a single first-order mode with eigenvalue

$$\lambda_{\text{brane}} = -H_b|_0. \quad (137)$$

Inside the generalized window (120), one has $H_b|_0 > 0$, so this mode decays.

The radion perturbation obeys

$$\delta\ddot{\phi} + 3H_b|_0 \delta\dot{\phi} + m_{\text{rad}}^2 \delta\phi = 0 \quad (138)$$

at zeroth order in the mixing. The corresponding characteristic equation is

$$\lambda^2 + 3H_b|_0 \lambda + m_{\text{rad}}^2 = 0, \quad (139)$$

with roots

$$\lambda_{\text{rad}}^\pm = -\frac{3H_b|_0}{2} \pm \frac{1}{2} \sqrt{9H_b^2|_0 - 4m_{\text{rad}}^2}. \quad (140)$$

For $H_b|_0 > 0$ and $m_{\text{rad}}^2 > 0$, both radion eigenvalues have non-positive real parts, with strictly negative real parts unless the system is exactly marginal. In the usual stabilized regime $m_{\text{rad}} \gg H_b|_0$, the eigenvalues are a damped pair,

$$\lambda_{\text{rad}}^\pm \simeq -\frac{3H_b|_0}{2} \pm i m_{\text{rad}}. \quad (141)$$

In this triangular leading-order system, the one-way curvature-induced mixing does not shift the eigenvalues. The constrained brane mode and radion mode are therefore separately stable when $H_b|_0 > 0$ and $m_{\text{rad}}^2 > 0$. The mixing condition (136) should be understood as the condition under which neglecting radion backreaction on the brane junction condition is self-consistent. A fully coupled stability result would require restoring that backreaction. To summarize, the constrained brane mode and the radion mode are separately stable throughout the generalized window

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}. \quad (142)$$

The special case $n_\rho = 0$ reproduces the simpler window $1 < c < \sqrt{2}$. Ordinary dust-like and radiation-like conserved matter, with $n_\rho \geq 2$, do not support the positive high-energy tracking root in this simplified force-balance model and require additional sources or a separate treatment.

C Toward a Controlled $O(\beta)$ Junction: The Ingoing-Vaidya Bulk

The brane equation of motion of Section 3 was treated as a phenomenological leading-order ansatz because the diagonal metric (8) is not an exact solution once μ is time-dependent. Here we set up the controlled calculation in the ingoing-Vaidya-AdS-Tangherlini bulk. The key outcome is that the *angular* junction condition—and hence the Friedmann constraint on which the entire tracking and stability analysis rests—is reproduced *exactly to all orders in β* , with the accretion corrections confined to the $\tau\tau$ condition and to a brane-bulk energy-exchange term.

C.1 Bulk geometry and null-dust source

In ingoing null (Eddington) coordinates (v, r, Ω_3) ,

$$ds_{(5)}^2 = -f(r, v) dv^2 + 2 dv dr + r^2 d\Omega_3^2, \quad f(r, v) = 1 - \frac{\mu(v)}{r^2} + k^2 r^2, \quad (143)$$

which, unlike (8), *is* an exact solution of $G_{MN} + \Lambda_5 g_{MN} = \kappa_5^2 T_{MN}$ for the ingoing null-dust source

$$T_{MN} = \rho_{\text{acc}}(v, r) \ell_M \ell_N, \quad \ell_M dx^M = -dv, \quad \ell_M \ell^M = 0, \quad (144)$$

provided

$$\kappa_5^2 \rho_{\text{acc}}(v, r) = \frac{3}{2} \frac{\mu'(v)}{r^3}, \quad \mu'(v) \equiv \frac{d\mu}{dv}, \quad (145)$$

i.e. the $D = 5$ form of (7) with $\mathcal{C}_5 = 2/3$. The accretion rate is first order: $\mu'(v)/\mu = O(\beta H_0)$.

C.2 Brane kinematics and the exact angular junction

Embedding the brane at $r = r_b(\tau)$ with $u^M = (\dot{v}, \dot{r}_b, 0, 0, 0)$, the normalization $u^M u_M = -1$ fixes

$$f \dot{v}^2 - 2 \dot{v} \dot{r}_b = 1 \quad \implies \quad \dot{v} = \frac{\dot{r}_b + \sqrt{\dot{r}_b^2 + f}}{f}. \quad (146)$$

The outward unit normal $n_M = (-\dot{r}_b, \dot{v}, 0, 0, 0)$ satisfies $n_M n^M = +1$ and $n_M u^M = 0$ identically, with radial component

$$\boxed{n^r = g^{rM} n_M = f\dot{v} - \dot{r}_b = \sqrt{f(r_b, v) + \dot{r}_b^2}}. \quad (147)$$

This is the central exact result: n^r has *precisely* the static- f form, with f now carrying the instantaneous $\mu(v)$. Hence the angular extrinsic curvature

$$K^i_j = \frac{n^r}{r_b} \delta_j^i = \frac{\sqrt{f(r_b, v) + \dot{r}_b^2}}{r_b} \delta_j^i \quad (148)$$

is identical in form to (20), and the $\mathbb{Z}_2$ angular Israel condition reproduces the modified Friedmann constraint (25) with $\mu \rightarrow \mu(v(\tau))$, *exactly and to all orders in β* . No $\dot{\mu}$ correction enters the constraint.

Consequence. The tracking solution of Section 3.3 and the constrained stability analysis of Appendix B are derived from this first-order constraint together with the brane conservation law. Since the constraint is exact, the leading-order tracking and stability results follow rigorously at zeroth adiabatic order; the $O(\beta)$ corrections flagged generically in (45) are now identified microscopically with the two effects below. This substantially reduces the phenomenological content: only the explicit accretion forcing in the second-order equation of motion, not the constraint or its leading consequences, ever required an ansatz.

C.3 The $\tau\tau$ condition and brane–bulk energy exchange

Two $O(\beta)$ effects distinguish the Vaidya $\tau\tau$ junction from the static one.

(i) *Closed form of the accretion correction to K^τ_τ .* The Christoffel symbols of (143) needed for the brane acceleration are

$$\Gamma_{vv}^v = \frac{1}{2}f_r, \quad \Gamma_{vv}^r = \frac{1}{2}(ff_r - f_v), \quad \Gamma_{vr}^r = -\frac{1}{2}f_r, \quad (149)$$

with $f_r \equiv \partial_r f$, $f_v \equiv \partial_v f = -\mu'(v)/r^2$, and $\Gamma_{vr}^v = \Gamma_{rr}^v = \Gamma_{rr}^r = 0$. Writing $W \equiv \sqrt{f + \dot{r}_b^2} = n^r$ and using $K_{\tau\tau} = -n_M a^M$ with $a^M = u^N \nabla_N u^M$, the proper-time component is

$$K^\tau_\tau = -K_{\tau\tau} = \dot{v} a^r - \dot{r}_b a^v. \quad (150)$$

The second derivative $\ddot{v}$ is eliminated through the τ -derivative of the normalization $f\dot{v}^2 - 2\dot{v}\dot{r}_b = 1$, which gives

$$\ddot{v} = \frac{\dot{v}(2\ddot{r}_b - \dot{f}\dot{v})}{2W}, \quad \dot{f} = f_r \dot{r}_b + f_v \dot{v}. \quad (151)$$

The resulting expression for K^τ_τ involves products of $\dot{v}$, $\dot{r}_b$, W , and f . These collapse under the exact identity

$$\dot{v}(W - \dot{r}_b) = 1, \quad (152)$$

which follows directly from the normalization (since $f\dot{v} - 2\dot{r}_b = 1/\dot{v}$ and $W = f\dot{v} - \dot{r}_b$). In particular, $\dot{v}^2(W - \dot{r}_b)^2 = 1$ eliminates all higher powers of $\dot{v}$ in favour of W and $\dot{r}_b$.

Separating the static ($f_v = 0$) and accretion ($f_v \neq 0$) parts: the static part reproduces (21),

$$K^\tau{}_\tau^{(0)} = \frac{\ddot{r}_b + \frac{1}{2}f_r}{W}, \quad (153)$$

(verified by checking that the $\ddot{r}_b$ coefficient is $\dot{v}(W - \dot{r}_b)/W = 1/W$ and the f_r terms combine to $f_r/(2W)$, both using (152)). The terms linear in f_v give

$$K_{\tau\tau}^{(1)} = \frac{f_v \dot{v}^3}{2} \cdot \frac{W - \dot{r}_b}{W} = \frac{f_v \dot{v}^2}{2W}, \quad (154)$$

where in the last step $\dot{v}^3(W - \dot{r}_b) = \dot{v}^2 \cdot [\dot{v}(W - \dot{r}_b)] = \dot{v}^2$ was used. Hence

$$\Delta K^\tau{}_\tau = -K_{\tau\tau}^{(1)} = -\frac{f_v \dot{v}^2}{2W} = \frac{\mu'(v) \dot{v}^2}{2r_b^2 \sqrt{f(r_b, v) + \dot{r}_b^2}}. \quad (155)$$

This is exact, linear in $\mu'(v)$, and manifestly first order in β .

Inserting $K^\tau{}_\tau = K^\tau{}_\tau^{(0)} + \Delta K^\tau{}_\tau$ into the $\mathbb{Z}_2$ Israel condition $2K^\tau{}_\tau = -\kappa_5^2(S^\tau{}_\tau - \frac{1}{3}S)$ and solving for the acceleration (the right-hand side is independent of μ' , since the brane stress-energy is intrinsic) gives

$$\ddot{r}_b = \ddot{r}_b^{\text{static}} - W \Delta K^\tau{}_\tau = \ddot{r}_b^{\text{static}} - \frac{\mu'(v) \dot{v}^2}{2r_b^2}. \quad (156)$$

It replaces the naive forcing $+\frac{\partial_t \mu}{2r_b^2} \dot{t}$ that a direct insertion into the diagonal static metric would assign. Since the physical accretion rate $d\mu/d\tau$ is reparametrization-invariant ($\partial_t \mu \dot{t} = \mu'(v) \dot{v}$), the phenomenological guess equals $+\mu' \dot{v}/(2r_b^2)$; the controlled result differs by a factor $-\dot{v}$ opposite sign and an extra redshift factor. The ansatz captured the parametric magnitude but not the coefficient or sign.

(ii) *Modified brane conservation from the bulk flux.* The ingoing null dust carries energy-momentum across the $\mathbb{Z}_2$ brane, so the worldvolume stress tensor is not separately conserved. The Codazzi projection of the bulk Einstein equation yields the energy equation

$$\dot{\rho}_{\text{br}} + 3H_b(\rho_{\text{br}} + p_{\text{br}}) = \mathcal{Q}, \quad \mathcal{Q} = -2T_{MN} n^M u^N|_{r_b}, \quad (157)$$

the factor of two being the $\mathbb{Z}_2$ jump. With $T_{MN} = \rho_{\text{acc}} \ell_M \ell_N$ and $\ell_M = -\partial_M v$, one computes $\ell \cdot u = \ell_v \dot{v} = -\dot{v}$ and $\ell \cdot n = \ell_v n^v = -\dot{v}$ (since $n^v = g^{vr} n_r = \dot{v}$), giving $T_{MN} n^M u^N = \rho_{\text{acc}} \dot{v}^2$. Using (145),

$$\mathcal{Q} = -2\rho_{\text{acc}} \dot{v}^2 = -\frac{3\mu'(v) \dot{v}^2}{\kappa_5^2 r_b^3} < 0 \quad (\mu' > 0). \quad (158)$$

The brane loses energy to the growing bulk black hole at first order in β .

C.4 Consistency and the first-order tracking problem

The two corrections (155) and (158) are not independent: the $\tau\tau$ junction, the exact Friedmann constraint, and the conservation law are tied by the contracted Gauss-Codazzi identity. To verify consistency, differentiate the high-energy Friedmann constraint (36)

using $H_b = \dot{r}_b/r_b$ and the modified conservation (158). Collecting terms and using $\dot{v}(W - \dot{r}_b) = 1$ to simplify, one recovers

$$\ddot{r}_b = r_b \left[(1 - n_\rho)B - \frac{\mu}{r_b^4} \right] - \frac{\mu'(v) \dot{v}^2}{2r_b^2}, \quad (159)$$

which is exactly the $\tau\tau$ result (156) with the static force identified as the zeroth-order acceleration (37). The identity $\dot{v}(W - \dot{r}_b) = 1$ is essential: it causes the $\mathcal{Q}$ contribution to $\dot{B}$ and the explicit $\mu'\dot{v}/r_b^4$ term from differentiating the dark radiation to combine into precisely $-\mu'\dot{v}^2/(2r_b^2)$. This constitutes an exact cross-check of the coefficient and sign of both (155) and (158).

Two consequences follow.

Zeroth-order robustness. What is genuinely robust is the *exact* angular constraint (147)–(25), and hence: the parametric scaling $R_h \sim (\beta H_0)^{-1}$ and $H_b \sim \beta H_0$; the $n = 1/2$ exponent (which depends only on the cutoff choice $\Lambda_{\text{IR}} = M_5^2 \sqrt{K_{\text{Weyl}}}$ and the swamp-land exponent, not on the precise value of c); and the *existence* of a high-energy tracking branch that is a quasi-static attractor ($H_b|_0 > 0$) over an $O(1)$ range of c . By contrast, the *precise* coefficient in the tracking relation (40), the precise window endpoints in (120), and the drift coefficient $\gamma(n_\rho)$ are *not* separately robust. All three are fixed by the quasi-static balance $\ddot{r}_b = 0$, which neglects the first-order Vaidya forcing, the brane–bulk energy exchange, and the kinematic terms $\ddot{c}$, $\dot{c}\dot{R}_h$, $\ddot{R}_h$, *all of the same order* $O(\beta^2 H_0^2)$ as the retained terms. Since $\gamma(n_\rho)$ is obtained by differentiating (40), it cannot be less robust than (40) itself; both receive the same order-unity relative corrections, fixed only once the first-order tracking problem is solved.

First-order problem. The forcing in (159) is $O(\beta H_0)$, the *same* order as the kinematic terms $\ddot{c}$, $\dot{c}\dot{R}_h$, $\ddot{R}_h$ neglected in the zeroth-order quasi-static condition and as the conservation source (158). A genuinely first-order tracking solution must retain all three simultaneously. With (155) and (158) now in closed form, the $O(\beta)$ problem is fully specified: it determines the first-order corrections to $c(t)$, to the drift coefficient $\gamma(n_\rho)$, and to $\dot{G}_4/G_4$. Carrying it out—and its numerical integration—is the concrete next step, no longer obstructed by the absence of a controlled bulk.

D Toward a 6D Completion: the Origin of $M_{\text{Pl}}^2 \propto R_{DD}$

The relation $M_{\text{Pl}}^2 = M_5^3 R_{DD}$ used in Section 5 could not be derived within the strictly 5D setup, whose only extra dimension is the noncompact radial direction r . Here we exhibit a six-dimensional completion in which the black-hole radial direction and the compact dark dimension coexist, and in which $M_{\text{Pl}}^2 \propto R_{DD}$ is *derived* by reduction. The reduction on the compact circle is carried explicitly; the radial integral, which fixes the proportionality constant, is set up and flagged.

D.1 Action and metric ansatz

Take the fundamental theory to be six-dimensional, with Planck mass M_6 (so the Einstein–Hilbert prefactor is M_6^4 , of mass dimension four):

$$S_6 = \frac{M_6^4}{2} \int d^6x \sqrt{-g_{(6)}} (R_{(6)} - 2\Lambda_6) + S_{\text{brane}} + S_{\text{bulk matter}}. \quad (160)$$

The coordinates are $x^M = (t, r, \Omega_3, y)$: four large directions (t, Ω_3) , the noncompact radial direction r carrying the Schwarzschild–Tangherlini black hole, and a compact circle $y \sim y + 2\pi$ of physical size R_{DD} . We take the warped product

$$ds_{(6)}^2 = g_{MN}^{(5)} dx^M dx^N + R_{DD}^2(x) dy^2, \quad (161)$$

with $g^{(5)}$ the 5D AdS–Tangherlini bulk (8) and $R_{DD}(x)$ depending only on the four large coordinates. The brane is a codimension-one hypersurface at $r = r_b(\tau)$ that *wraps* the circle y ; its worldvolume is (τ, Ω_3, y) , with induced line element $-d\tau^2 + r_b^2 d\Omega_3^2 + R_{DD}^2 dy^2$. Because the (t, r, Ω_3) block is untouched by (161), the $\mathbb{Z}_2$ Israel junction in r is exactly that of Section 3: the circle y is a spectator worldvolume direction that rescales the effective brane tension by $2\pi R_{DD}$. The Friedmann constraint, tracking solution, and constrained stability are then expected to carry over, *provided* the wrapped-brane junction is verified: the yy component of the Israel condition and the $2\pi R_{DD}$ tension rescaling—with the 6D trace subtraction $\frac{1}{D-2} = \frac{1}{4}$ replacing the 5D value $\frac{1}{3}$ —must be checked, since a codimension-one brane wrapping y does not automatically obey the 5D trace projection. We assume this holds; the explicit verification is part of the content flagged in the caveat below.

D.2 Reduction on the dark circle

For the warped product (161),

$$\sqrt{-g_{(6)}} = R_{DD} \sqrt{-g_{(5)}}, \quad R_{(6)} = R_{(5)} - \frac{2}{R_{DD}} \square_{(5)} R_{DD}, \quad (162)$$

in exact analogy with (89). Since $R_{DD} = R_{DD}(x)$ depends only on the four large coordinates, $\square_{(5)} R_{DD}$ is a total derivative under integration. Performing $\int_0^{2\pi} dy$,

$$\frac{M_6^4}{2} \int d^6 x \sqrt{-g_{(6)}} R_{(6)} = \frac{2\pi M_6^4}{2} \int d^5 x \sqrt{-g_{(5)}} R_{DD}(x) R_{(5)} + (\text{bndy}), \quad (163)$$

which is the 5D Einstein–Hilbert action (3) with a field-dependent effective coupling

$$\boxed{M_{5,\text{eff}}^3(x) = 2\pi M_6^4 R_{DD}(x) .} \quad (164)$$

This reduction is exact for the ansatz (161).

D.3 4D Planck mass

The 4D Planck mass follows from integrating the 5D action over the noncompact radial direction. With $\sqrt{-g_{(5)}} = r^3 \sqrt{\hat{g}_{S^3}}$ (the lapse and radial metric factors cancel, $\sqrt{g_{tt} g_{rr}} = 1$),

$$M_{\text{Pl}}^2 = M_{5,\text{eff}}^3 \ell_r = 2\pi M_6^4 \ell_r R_{DD}, \quad \ell_r \equiv \int dr w(r), \quad (165)$$

with $w(r)$ the radial measure of the reduced geometry. Because the ansatz (161) makes $g^{(5)}$ independent of R_{DD} , the radial volume ℓ_r is independent of R_{DD} , and the R_{DD} -dependence is strictly linear:

$$\boxed{M_{\text{Pl}}^2 \propto R_{DD}, \quad G_4 = \frac{1}{8\pi M_{\text{Pl}}^2} \propto \frac{1}{R_{DD}} .} \quad (166)$$

Identifying the constant combination $M_5^3 \equiv 2\pi M_6^4 \ell_r$ reproduces the relation $M_{\text{Pl}}^2 = M_5^3 R_{DD}$ assumed in Section 5, with R_{DD} unambiguously the size of a compact dimension distinct from the radial r . This reduces the assumption of Section 5 to the conditions on ℓ_r and the wrapped-brane junction stated below; it becomes a derivation only once those conditions are established, which we do not do here. The drift $\dot{G}_4/G_4 = -\dot{R}_{DD}/R_{DD}$, and hence the phenomenology of Section 6, follow whenever ℓ_r is static.

Caveat on ℓ_r . Equation (165) is schematic, and the measure $w(r)$ must be identified with care. The factor r^3 in $\sqrt{-g_{(5)}}$ is the volume element of the three-sphere, which is part of 4D worldvolume (τ, Ω_3) rather than of the radial reduction measure; the genuine $w(r)$ is the *dimensionless* normalized graviton zero-mode (warp) profile along r , in direct analogy with the factor $e^{-2k|y|}$ of Randall–Sundrum, so that $\ell_r = \int dr w(r)$ has dimension [length] and $M_{\text{Pl}}^2 = M_{5,\text{eff}}^3 \ell_r$ carries the correct [mass]². As in RS-II, normalizability of the zero mode requires the radial volume to be regulated by the AdS warp factor or a second brane / infrared cutoff; in the weak-AdS regime adopted in the main text ($k \ll \beta H_0$, Eq. (61)) the warping is mild and this regularization is delicate. A complete treatment must specify the radial cutoff, compute the normalized zero-mode measure $w(r)$, verify that ℓ_r is finite and approximately static (so the time dependence of G_4 is carried by $R_{DD}(t)$ alone), and check the wrapped-brane junction with the $2\pi R_{DD}$ tension rescaling. This is the content still required to promote (165) to a fully controlled statement.

D.4 Robustness of the $n = 1/2$ scaling

The black hole resides in the 5D factor, so the leading Weyl invariant of the 6D geometry is still $K_{\text{Weyl}} \propto \mu^2/r^8$ up to corrections suppressed by the slow variation of R_{DD} . The curvature cutoff must, however, be built from a *constant* mass scale. The field-dependent effective coupling $M_{5,\text{eff}}^3 = 2\pi M_6^4 R_{DD}$ (Eq. (164)) is *not* admissible here: using $M_{5,\text{eff}}^2 \propto R_{DD}^{2/3}$ would give $\Lambda_{\text{IR}} \propto R_{DD}^{2/3} c^{-4} R_h^{-2}$, and solving $R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}$ self-consistently would yield $R_{DD} \propto c^{6/7} R_h^{3/7}$ (i.e. $n = 3/7$), altering the exponent. The cutoff is instead constructed from the constant fundamental scale M_6 (mass dimension one), so that

$$\Lambda_{\text{IR}} = M_6^2 \sqrt{K_{\text{Weyl}}} \propto c^{-4} R_h^{-2}, \quad R_{DD,\text{min}} \sim \Lambda_{\text{IR}}^{-1/4} \propto c R_h^{1/2}, \quad (167)$$

which is dimensionally correct ($[M_6^2 \sqrt{K_{\text{Weyl}}}] = [\text{mass}]^4$) and reproduces $n = 1/2$. Equivalently one may use the static combination $M_5^3 \equiv 2\pi M_6^4 \ell_r$, whose M_5 is constant when ℓ_r is static; the essential point is that the scale entering Λ_{IR} must be the field-independent one, not $M_{5,\text{eff}}$. With this proviso the exponent $n = 1/2$ is unaffected by the embedding, as anticipated in the main text.

E Glossary of Principal Symbols and Relations

β : Dimensionless slow-accretion parameter, $\beta \equiv \frac{\dot{M}}{MH_0}$. Here $\dot{M}$ is the accretion rate (how fast the black hole is gaining mass), M is the black hole’s total (ADM) mass, and H_0 is the present-day Hubble constant. The condition $\beta \ll 1$ means the black hole grows extremely slowly on cosmological timescales.

$R_h(t)$: Horizon radius of the 5D bulk black hole. It sets the characteristic curvature scale of the black hole and is related to the mass parameter by $R_h^2 \simeq \mu$.

$\mu(t)$: Mass parameter appearing in the 5D metric (dimensions of length squared). It is directly proportional to the physical mass M of the black hole and grows slowly because of accretion.

$r_b(t)$: Radial coordinate of the three-brane in the 5D bulk. This is the dynamical position of our 4D universe inside the extra dimension.

$c(t)$: Dimensionless tracking parameter defined by the relation $r_b(t) = c(t)R_h(t)$. It varies slowly with time and controls how far outside the black-hole horizon the brane sits.

$\Lambda_{\text{IR}}(t)$: Infrared cutoff (an energy-density scale with dimensions $[\text{mass}]^4$) that appears in the effective 4D radion potential. It is supplied dynamically by the local curvature of the bulk black hole.

$K_{\text{Weyl}}(r, t)$: Weyl part of the Kretschmann scalar (a curvature invariant). It measures the tidal gravitational field due to the black hole alone (the pure “Weyl” piece that survives after subtracting the constant anti-de Sitter background).

M_5 : 5D Planck mass. It sets the strength of gravity in the full 5D theory (analogous to the 4D Planck mass M_{Pl}).

κ_5^2 : 5D gravitational coupling constant, $\kappa_5^2 \equiv M_5^{-3}$.

σ : Brane tension (energy per unit area on the three-brane). It is fine-tuned against the bulk cosmological constant in the Randall-Sundrum construction.

k : Anti-de Sitter curvature scale in the bulk, related to the cosmological constant and brane tension by $k = \kappa_5^2 \sigma / 6$.

$\rho_{\text{br}}, p_{\text{br}}$: Energy density and pressure of ordinary matter and radiation on the brane (the 4D “visible” universe).

$n_\rho = 3(1 + w)$: Effective equation-of-state index for the dominant brane matter component, where $w = p_{\text{br}}/\rho_{\text{br}}$. It enters the stability window for the tracking solution.

$R_{DD}(t)$: Size (compactification radius) of the dark (mesoscopic) extra dimension. This is the modulus field we are trying to stabilize.

$\phi(x)$: Canonically normalized radion field (the 4D scalar that describes fluctuations of R_{DD}).

$G_4(t)$: Effective 4D Newton constant. Because the extra dimension is dynamical, G_4 inherits a slow time dependence: $G_4(t) \propto 1/R_{DD}(t)$.

H_b : Hubble parameter measured on the brane (the expansion rate of our 4D universe).

Φ : The 5D bulk scalar field used in the Goldberger-Wise stabilization mechanism. It has a bulk mass m_{GW} and couples to the visible brane through a quartic potential with vacuum expectation value v_{br} .

m_{GW} : The mass parameter of the bulk Goldberger-Wise scalar Φ in the 5D theory. It controls how strongly the scalar varies across the extra dimension and helps generate the radion potential.

v_{br} : The vacuum expectation value (vev) that the Goldberger-Wise scalar Φ takes on the visible brane. It is one of the parameters that sets the minimum of the radion potential.

c_{Cas} : Numerical coefficient appearing in the Casimir energy contribution $V_{\text{Cas}} \propto -c_{\text{Cas}}/R_{DD}^4$ to the effective radion potential. It encodes the effect of quantizing the bulk graviton (and other fields) on the compact

extra dimension.

ξ : Dimensionless coupling constant that controls the strength of the non-minimal curvature interaction $\frac{\xi}{M_5^2} K_{\text{Weyl}} \Phi^2$ between the bulk scalar and the Weyl curvature of the black hole

f_ϕ : Decay constant (normalization scale) of the canonically normalized radion field ϕ . It sets the overall strength with which the radion couples to matter and gravity in the 4D effective theory.

λ_{br} : Quartic self-coupling strength in the brane-localized potential $V_{\text{br}}(\Phi) = \frac{\lambda_{\text{br}}}{4} (\Phi^2 - v_{\text{br}}^2)^2$. In the limit $\lambda_{\text{br}} \rightarrow \infty$ the scalar is fixed to its vacuum expectation value on the brane

$f(r, t)$: The metric lapse function in the 5D Schwarzschild-Tangherlini geometry. It encodes gravitational redshift and determines the proper radial distance in the bulk.

$\dot{M}$: Accretion rate of the black hole (time derivative of its ADM mass M). It appears in the definition of the slow-accretion parameter β .

$\varepsilon(t)$: Small dimensionless perturbation around the adiabatic tracking solution, $r_b(t) = c(t)R_h(t)(1 + \varepsilon(t))$, used in the linear stability analysis.

$\Gamma(t)$: Positive decay rate of perturbations $\varepsilon(t)$ in the tracking solution, equal to the background Hubble rate evaluated on the attractor, $\Gamma = H_b|_0$ (so $\dot{\varepsilon} = -H_b|_0 \varepsilon$). It guarantees that deviations from the attractor $r_b = cR_h$ are exponentially damped inside the window $1 < c^2 < \frac{2-n_\rho}{1-n_\rho}$.

$\delta\phi$: Small fluctuation of the canonically normalized radion field $\phi(x)$ around its moving minimum $\phi_{\text{min}}(t)$.

m_{rad} : Instantaneous mass of the radion field evaluated at the dynamical minimum of the effective potential. The condition $m_{\text{rad}} \gg H_0$ ensures the radion adiabatically tracks the slowly moving minimum.

η_{ad} : Adiabaticity parameter measuring how slowly the minimum $R_{DD, \text{min}}(t)$ moves compared to the radion oscillation frequency; the tracking is valid when $\eta_{\text{ad}} \ll 1$.

$F(r_b, \dot{r}_b)$: Kinematic function that encodes the effect of stress-energy conservation ($\dot{\rho}_{\text{br}} + 3H_b(\rho_{\text{br}} + p_{\text{br}}) = 0$) in the second-order brane equation of motion.

$V_{\text{eff}}^{\text{brane}}(r_b, t)$: Effective potential governing the mechanical motion of the brane radius $r_b(\tau)$ after the Randall-Sundrum cancellation has been performed.

$\dot{t}$: Redshift factor that converts derivatives with respect to bulk coordinate time t into derivatives with respect to proper time τ on the brane: $\dot{t} = \sqrt{f(r_b, t) + \dot{r}_b^2}/f(r_b, t)$.

a^b : Four-acceleration of the brane's four-velocity u^a . It is purely normal to the brane and appears when computing the extrinsic curvature component K_τ^τ .

a^r : Radial component of the brane four-acceleration; explicitly $a^r = \ddot{r}_b + f'/2$, where $f' = \partial_r f$.

$h^{\tau\tau}$: Component of the projector onto the brane; equals -1 in the proper-time gauge used throughout the paper.

A : Low-energy coefficient in the modified Friedmann equation, $A = \kappa_5^4 \sigma \rho_{\text{br}}/18$. On the tracking background the conserved-matter force balance fixes it to $A = \frac{2}{(2-n_\rho)} c^{-4} R_h^{-2}$ (reducing to $c^{-4} R_h^{-2}$ only in the vacuum-like case $n_\rho = 0$).

- $\boxed{\mathcal{G}(c, n_\rho)}$: Dimensionless factor that appears in the perturbation decay rate: $\Gamma = H_b \mathcal{G}(c, n_\rho)$, with explicit form $\mathcal{G} = (n_\rho + 4 - 2c^2)/[2(2 - c^2)]$.
- $\boxed{\lambda_{\phi, \pm}}$: Eigenvalues of the radion fluctuation mode in the linearized stability analysis: $\lambda_{\phi, \pm} = -3H_b|_0/2 \pm \sqrt{(3H_b|_0/2)^2 - m_{\text{rad}}^2}$.
- $\boxed{\mathbb{Z}_2}$: Discrete reflection (orbifold) symmetry imposed across the brane; it doubles the extrinsic curvature jump in the Israel junction conditions.
- $\boxed{\Sigma}$: The three-brane world-volume (the 4-dimensional hypersurface on which our universe lives).
- $\boxed{R_0}$: Arbitrary fiducial reference scale used to define the canonically normalized radion field via $R_{DD}(x) = R_0 \exp(\phi(x)/f_\phi)$.
- $\boxed{\beta_\phi}$: Dimensionless coupling constant of the radion to matter density in the chameleon screening mechanism.
- $\boxed{p}$: Positive exponent that controls the density dependence of the local radion mass: $m_{\text{rad, loc}}^2 \propto (\rho_m/\rho_{\text{cos}})^p$.
- $\boxed{Y_p}$: Primordial helium-4 mass fraction; its sensitivity to a drifting G_4 is used to constrain integrated changes in the effective Newton constant.
- $\boxed{T_{\text{BBN}}}$: Temperature at Big Bang Nucleosynthesis; appears in estimates of the integrated drift of G_4 between BBN and today.
- $\boxed{m_{KK}}$: Mass of the lightest Kaluza-Klein graviton mode, approximately $1/R_{DD}(t)$; its slow evolution is a potential observational signature.
- $\boxed{c_{\text{dS}}, c'_{\text{dS}}}$: Positive constants appearing in the refined de Sitter swampland conjectures that bound the slope and curvature of positive-energy potentials.
- $\boxed{\bar{\beta}}$: Time-averaged value of the slow-accretion parameter; used when integrating the drift of G_4 or R_h over cosmological history.

F Glossary of Labeled Equation

F.1 Equations from the Main Text

$$\beta \equiv \frac{\dot{M}}{MH_0}$$

Dimensionless slow-accretion parameter. It is the fundamental small parameter controlling the adiabatic approximation.

$$R_{DD} \sim \Lambda_{\text{IR}}^{-1/4}$$

Dark-dimension swampland scaling relation that determines the compactification radius from the infrared cutoff.

$$ds_{(5)}^2 = -f(r, t) dt^2 + f(r, t)^{-1} dr^2 + r^2 d\Omega_3^2, \quad f(r, t) = 1 - \frac{\mu(t)}{r^2} + k^2 r^2$$

Five-dimensional AdS–Schwarzschild–Tangherlini metric with time-dependent mass parameter.

$$K_{\text{Weyl}}(r, t) = \frac{72\mu(t)^2}{r^8}$$

Weyl part of the Kretschmann scalar used to construct the dynamical infrared cutoff.

$$[K_{\mu\nu}] = -\kappa_5^2 \left(S_{\mu\nu} - \frac{1}{3} S h_{\mu\nu} \right)$$

Israel junction conditions with $\mathbb{Z}_2$ symmetry.

$$H_b^2 = \frac{\kappa_5^4}{36} \rho_{\text{br}}^2 + \frac{\kappa_5^4 \sigma \rho_{\text{br}}}{18} + \frac{\mu(t)}{r_b^4} - \frac{1}{r_b^2}$$

Modified Friedmann equation on the brane.

$$\dot{r}_b = \dot{c} R_h + c \dot{R}_h$$

First kinematic relation obtained by differentiating the tracking ansatz.

$$\ddot{r}_b = \ddot{c} R_h + 2\dot{c} \dot{R}_h + c \ddot{R}_h$$

Second kinematic relation obtained by differentiating the tracking ansatz.

$$\frac{1}{R_h} \frac{dR_h}{d\tau} = \dot{t} \cdot \frac{1}{2} \beta H_0$$

Evolution of the horizon radius expressed in brane proper time.

$$r_b(t) = c(t) R_h(t)$$

Central tracking ansatz.

$$\frac{\ddot{r}_b}{r_b} = (1 - n_\rho)B - \frac{\mu}{r_b^4}, \quad B \equiv \frac{\kappa_5^4 \rho_{\text{br}}^2}{36}$$

High-energy brane acceleration equation.

$$c^4(t) = \frac{36}{(1 - n_\rho) \kappa_5^4 R_h^2(t) \rho_{\text{br}}^2(t)}$$

Tracking relation determining $c(t)$ in the high-energy regime.

$$n_\rho < 1, \quad 1 < c^2 < \frac{2 - n_\rho}{1 - n_\rho}$$

Generalized stability window for the tracking solution.

$$r_b(t) = c(t) R_h(t) (1 + \varepsilon(t))$$

Linear perturbation about the tracking solution.

$$\dot{\varepsilon} = -H_b|_0 \varepsilon + O(\beta H_0 \varepsilon)$$

Evolution equation for perturbations around the tracking solution.

$$H_b^2|_0 = \frac{1}{c^4 R_h^2} \left(\frac{2 - n_\rho}{1 - n_\rho} - c^2 \right)$$

Background Hubble rate on the tracking solution.

$$\Lambda_{\text{IR}}(t) \equiv M_5^2 \sqrt{K_{\text{Weyl}}(r_b, t)} \propto c(t)^{-4} R_h(t)^{-2}$$

Curvature-set infrared cutoff.

$$R_{DD, \text{min}}(t) \propto c(t) R_h(t)^{1/2}$$

Dynamical scaling of the compactification radius.

$$G_4(t) = \frac{1}{8\pi M_5^3 R_{DD}(t)}$$

Time-dependent effective 4D Newton constant.

$$\frac{\dot{G}_4}{G_4} = -\frac{\dot{c}}{c} - \frac{1}{4}\dot{t}\beta H_0$$

Cosmological drift rate of the effective Newton constant.

F.2 Equations from Appendix A

$$H_b^2 = \frac{\kappa_5^4}{36}\rho_{\text{br}}^2 + \frac{\kappa_5^4\sigma\rho_{\text{br}}}{18} + \frac{\mu}{r_b^4} - \frac{1}{r_b^2}$$

Modified Friedmann equation after imposing the angular Israel condition.

$$\frac{\ddot{r}_b}{r_b} = (1 - n_\rho)B - \frac{\mu}{r_b^4}$$

High-energy brane acceleration equation at zeroth adiabatic order.

F.3 Equations from Appendix B

$$\dot{\varepsilon} = -H_b|_0 \varepsilon$$

Exact decay law for the constrained brane perturbation mode.

$$\lambda_{\text{brane}} = -H_b|_0, \quad \lambda_{\text{rad}}^\pm = -\frac{3H_b|_0}{2} \pm \frac{1}{2}\sqrt{9H_b^2|_0 - 4m_{\text{rad}}^2}$$

Eigenvalues of the coupled brane–radion system.

$$\delta\ddot{\phi} + 3H_b|_0\delta\dot{\phi} + m_{\text{rad}}^2\delta\phi = \mathcal{S}_{\phi\varepsilon}\varepsilon$$

Perturbation equation for the radion with curvature-induced mixing.

F.4 Equations from Appendix C: Ingoing-Vaidya

$$ds_{(5)}^2 = -f(r, v) dv^2 + 2 dv dr + r^2 d\Omega_3^2$$

Ingoing-Vaidya form of the 5D bulk metric.

$$\dot{v} = \frac{\dot{r}_b + \sqrt{\dot{r}_b^2 + f}}{f}$$

Expression for the advanced-time derivative on the brane.

$$n^r = \sqrt{f(r_b, v) + \dot{r}_b^2}$$

Radial component of the outward unit normal in Vaidya coordinates.

$$\Delta K^\tau{}_\tau = \frac{\mu'(v)\dot{v}^2}{2r_b^2\sqrt{f + \dot{r}_b^2}}$$

Closed-form $O(\beta)$ correction to $K^\tau{}_\tau$ from accretion.

$$\ddot{r}_b = \ddot{r}_b^{\text{static}} - \frac{\mu'(v)\dot{v}^2}{2r_b^2}$$

Controlled accretion forcing term in the brane acceleration.

$$\mathcal{Q} = -\frac{3\mu'(v)\dot{v}^2}{\kappa_5^2 r_b^3}$$

Brane–bulk energy exchange term due to null-dust flux.

F.5 Equations from Appendix D: 6D Completion

$$ds_{(6)}^2 = g_{MN}^{(5)} dx^M dx^N + R_{DD}^2(x) dy^2$$

Warped-product metric ansatz for the six-dimensional completion.

$$M_{5,\text{eff}}^3(x) = 2\pi M_6^4 R_{DD}(x)$$

Field-dependent effective 5D Planck mass after reduction over the compact dark circle.

$$M_{\text{Pl}}^2 = 2\pi M_6^4 \ell_r R_{DD}$$

Relation between the 4D Planck mass and the compactification radius in the 6D setup.

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